

Lottery Prior: Randomized Neural Compression for Zero-Shot Inverse Problems

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Abstract

We study zero-shot inverse problems, where a clean signal is recovered from a single degraded observation without external training data. Contrary to the common belief that such problems require highly complex models, we show that a lightweight neural network, when combined with entropy and complexity regularization in a compression-based formulation, is sufficient for high-quality restoration. We propose Lottery Prior, a compression-based inverse solver that leverages architectural priors from random networks and induces a family of implicit priors through randomness, enabling ensemble-based refinement. We further derive non-asymptotic error bounds for compression-based maximum-likelihood inverse solvers, revealing how rate-distortion constraints act as implicit regularizers. Experiments on denoising, noisy super-resolution, and inpainting demonstrate that our method achieves state-of-the-art with significantly fewer effective parameters. Project page: <https://eedavidwu.github.io/LotteryPrior/>

1. Introduction

In an inverse problem, one aims to recover an unknown signal $\mathbf{x}$ from a degraded observation $\mathbf{y} = \mathcal{A}(\mathbf{x}, \mathbf{z})$, where $\mathcal{A}(\cdot)$ denotes a degradation operator and $\mathbf{z}$ is the noise component. Many inverse problems, such as denoising, super-resolution, and inpainting, can be formulated as energy minimization problems (see Fig. 2):

$$\mathbf{P1:} \quad \mathbf{x}^* = \arg \min_{\hat{\mathbf{x}}} E(\hat{\mathbf{x}}, \mathbf{y}) + \lambda R(\hat{\mathbf{x}}), \quad (1)$$

where $E(\hat{\mathbf{x}}, \mathbf{y})$ is a task-dependent fidelity term, and $R(\hat{\mathbf{x}})$

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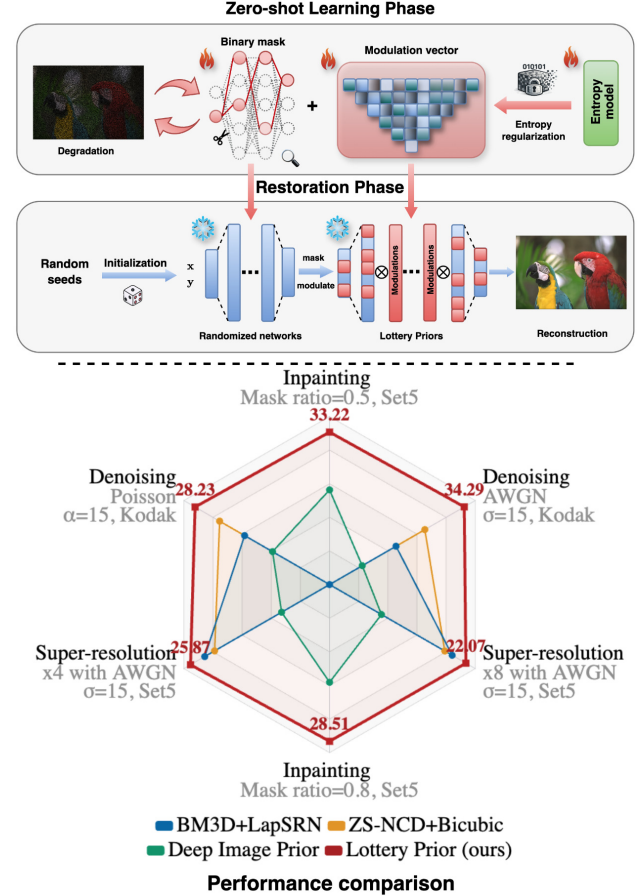


Figure 1. Up: Illustration of Lottery Priors for zero-shot inverse problems. During the learning phase, the degraded observation is fitted with a neural network regularized by a binary mask and entropy-constrained latent modulations. At restoration, a modulated subnetwork of a randomly initialized network reconstructs the image. Varying mask ratio and random initialization induce a family of lottery priors, enabling further ensemble-based refinement. **Down:** Performance comparison of different tasks, where Lottery Prior achieves state-of-the-art performance.

is a regularizer with hyper-parameter λ that captures prior knowledge about the generic structure of the source.

Conventional inverse problem solvers rely on handcrafted models tailored to the forward operator, but can produce unnatural solutions in ill-posed settings (Hegde, 2018). Recent advances in representation learning, including genera-

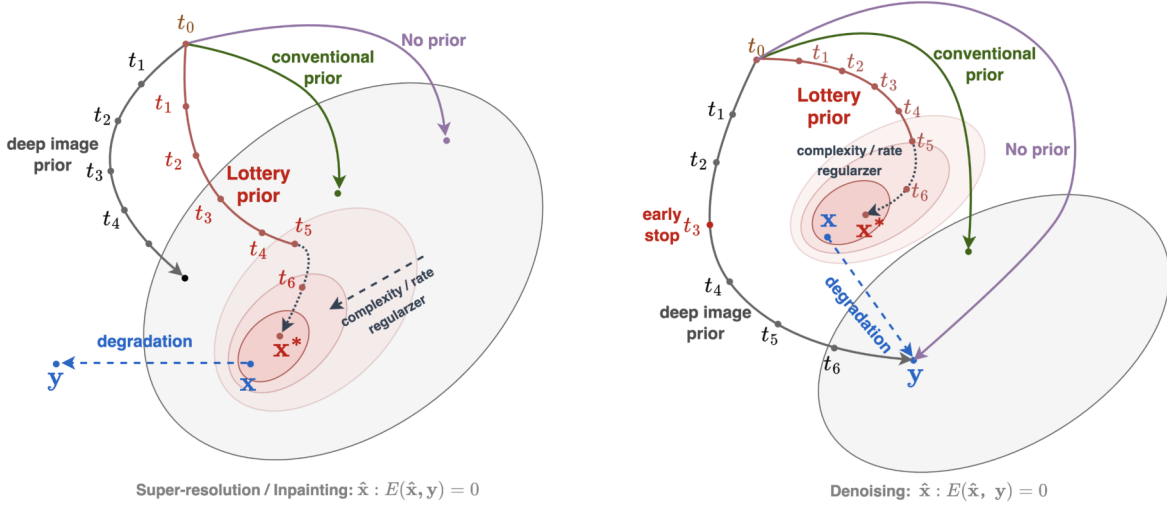


Figure 2. Inverse problem with priors. We consider reconstructing $\mathbf{x}$ from an observation $\mathbf{y}$. **Left:** For problems such as super-resolution or inpainting, multiple solutions satisfy $E(\hat{\mathbf{x}}, \mathbf{y}) = 0$ (gray region with null energy). Without a strong prior, optimization may converge far from the ground truth. Conventional priors and DIP guide the solution toward $\mathbf{x}$, while the proposed *lottery prior* further regularizes this solution via rate and complexity constraints. **Right:** For denoising, the ground truth has non-zero cost $E(\hat{\mathbf{x}}, \mathbf{y}) \approx \sigma^2$, where σ^2 is the noise variance. While DIP can reach low-energy solutions with extensive optimization, it tends to overfit and requires early stopping to recover $\mathbf{x}$. In contrast, *lottery priors* provide additional regularization that mitigates overfitting without early stopping. Moreover, inherent randomness induces multiple solutions near the ground truth, enabling ensemble-based refinement in both cases.

tive models (Asim et al., 2020), transformers (Zamir et al., 2022), and diffusion models (Chung et al., 2023), have substantially improved reconstruction performance by learning powerful priors. However, despite their success on standard benchmarks, these methods face significant practical challenges in terms of *data efficiency* and *deployment complexity* (Qayyum et al., 2022). They typically require large-scale paired training data and computationally intensive models, and are sensitive to distribution shifts, often necessitating retraining and data recollection, which limits practical deployment. To reduce reliance on paired data, self-supervised methods learn restorations directly from noisy observations without clean ground truth (Soltanayev & Chun, 2018; Batson & Royer, 2019). However, they generally require large noisy datasets, which can be costly or impractical to collect in scenarios such as medical imaging or remote sensing. This has motivated growing interest in zero-shot inverse solvers, which recover clean signals from a single degraded observation by enforcing measurement consistency while favoring solutions on the natural signal manifold.

A notable example is compression-based zero-shot denoising, which naturally instantiates this principle by exploiting the greater compressibility of clean signals relative to noise (Donoho, 2002). By restricting reconstructions to a codebook, compression induces a structure prior without external training. Formally, lossy compression optimizes a rate–distortion (RD) trade-off (Cover, 1999):

$$\mathbf{P2:} \quad \min D(\hat{\mathbf{x}}, \mathbf{x}) + \lambda R, \quad (2)$$

where $D(\cdot)$ denotes a distortion measure and $R(\cdot)$ is the

corresponding rate term, with λ controlling the trade-off. Viewed through Eq. (1), and taking the noisy observation $\mathbf{y}$ as the compression target, Eq. (2) can be interpreted as a zero-shot denoising solver, with the rate term acting as an implicit regularizer that favors structured, natural solutions. Recent work (Zafari et al., 2026) further demonstrates this potential by training neural compression models directly on patches from a single noisy image. However, this autoencoder-based and patch-driven approach introduces substantial complexity and limits the use of global image context, which often leads to suboptimal reconstructions. Moreover, without additional priors, such methods are largely limited to denoising and do not readily extend to more general inverse problems.

Contrary to the common belief that zero-shot inverse problems require complex models to learn multi-scale features, we show that a lightweight neural network with entropy constraints and flexible compression parameterizations is sufficient for high-quality restoration. This suggests the potential for an inverse solver that is simultaneously *good* (reconstruction quality), *fast* (inference speed), and *cheap* (restoration mechanism), exemplifying a clear “*less is more*” paradigm. Concretely, our approach is motivated by two observations: (i) untrained neural networks inherently encode useful structural priors through their architectures, as evidenced by deep image prior (DIP) (Ulyanov et al., 2020), and (ii) random networks can serve as low-cost, flexible implicit codecs (Wu et al., 2025).

Building on these insights, we propose *Lottery Prior*, a

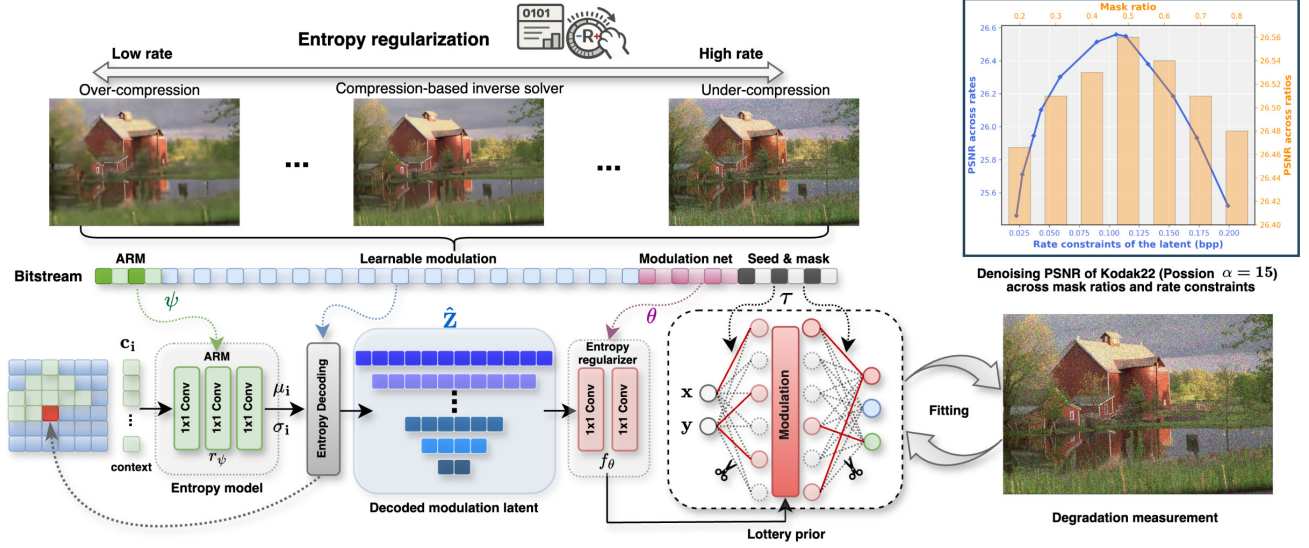


Figure 3. Illustration of the image restoration process in *Lottery Prior*. Each degraded observation is fitted using an implicit codec parameterized by a learnable latent and a subnetwork of a randomized neural network. An entropy model imposes rate constraints on the latent, enabling a well-calibrated rate–distortion trade-off for effective restoration. At inference, the binary mask and random initialization seed configure the synthesis network, while latent-driven modulations guide the image reconstruction. Empirically, a mask ratio of 50% provides a strong architectural prior with favorable statistical encoding capacity and improved restoration performance.

randomized neural compression-based inverse solver that combines architectural priors with entropy and complexity regularization. The inherent randomness induces a family of implicit priors whose ensemble can further enhance restoration performance. Specifically, this is formulated as:

$$\mathbf{P3:} \quad (\mathbf{Z}^*, \tau^*) = \arg \min_{\mathbf{Z}, \tau} \left[\underbrace{D(\mathcal{A}(g_{\mathbf{w}_0 \odot \tau}(\mathbf{Z})), \mathbf{y})}_{\text{Measurement alignment}} + \underbrace{\lambda R(\mathbf{Z})}_{\text{Entropy regularizer}} + \underbrace{\beta J(\tau)}_{\text{Complexity regularizer}} \right], \quad (3)$$

with $\hat{\mathbf{x}} = g_{\mathbf{w}_0 \odot \tau^*}(\mathbf{Z}^*)$,

where $g_{\mathbf{w}_0 \odot \tau}$ denotes a randomly initialized neural network with parameters $\mathbf{w}_0$ masked via a binary vector $\tau \in \{0, 1\}^{|\mathbf{w}_0|}$. The network g reconstructs the signal from the latent $\mathbf{Z}$, which is subject to an entropy constraint $R(\mathbf{Z})$, while $J(\tau)$ enforces a sparsity-based complexity constraint on the architecture. Here, entropy and complexity regularization act as implicit priors, where mask optimization plays a DIP-like role, and different random initializations induce an ensemble of solutions for improved reconstruction.

We show that this formulation is highly competitive across various zero-shot inverse problems, including denoising, noisy super-resolution, and inpainting (see Fig. 1). Notably, model complexity is tightly controlled by rate and structural priors, resulting in significantly fewer effective parameters than prior approaches. *To the best of our knowledge, this is the first work to explicitly study the priors induced by low-cost implicit codecs for various inverse problems.* These

findings open a promising direction toward lightweight plug-and-play inverse solvers grounded in compression-based principles. The main contributions are:

- We propose *Lottery Prior*, a zero-shot inverse solver that unifies random-network architectural priors with entropy and complexity regularization, together inducing a family of implicit priors with ensemble-like behavior, thereby improving the restoration performance.
- We derive non-asymptotic measurement- and signal-domain error bounds for compression-based maximum-likelihood inverse solvers, showing how rate–distortion constraints act as implicit regularizers with provable guarantees, including denoising as a special case.
- Extensive experiments show that our method achieves state-of-the-art performance on various zero-shot inverse problems such as denoising, noisy super-resolution, and inpainting, while using significantly fewer effective parameters.

2. Related work

Neural data compression. Neural data compression methods can be broadly categorized into two paradigms: AE-based and implicit codecs. AE-based codecs replace hand-crafted transforms with learned auto-encoders (Ballé et al., 2018; Jiang et al., 2023), achieving strong RD performance but requiring extensive training and offering limited zero-shot flexibility. In contrast, implicit codecs optimize a para-

metric function per signal and encode its parameters, enabling efficient decoding. Representative methods, such as COIN++ (Dupont et al., 2022), COMBINER (He et al., 2024), COOL-CHIC family (Leguay et al., 2023; Kim et al., 2024; Ladune et al., 2026), LotteryCodec (Wu et al., 2025), and MoRIC (Li et al., 2026) achieve strong RD performance with efficient decoding. Together, these works highlight the rapid progress of neural compression toward practical codecs that jointly optimize RD performance and efficiency.

Zero-shot inverse problem. Zero-shot inverse problems aim to recover a clean signal from a single degraded observation without any external training data or pretrained source knowledge. Untrained priors, such as DIP (Ulyanov et al., 2020) and Deep Decoder (DD) (Heckel & Hand, 2019), exploit architectural bias and have been applied to various inverse problems. However, most zero-shot work focuses on denoising, including single-image adaptations of self-supervised methods (Batson & Royer, 2019; Quan et al., 2020) and compression-based denoising, which leverages the higher compressibility of structured signals over noise (Donoho, 2002; Weissman & Ordentlich, 2005). While recent zero-shot neural codecs (Zafari et al., 2026) train directly on the observed signal, they remain AE-based and patch-driven, limiting the use of architectural priors, global context, and extension beyond denoising. Dataset-free formulations have also been explored for inverse problems such as compressive sensing and blind super-resolution (Pang et al., 2020; Wang et al., 2022; Shocher et al., 2018; Yue et al., 2022), but remain comparatively underexplored. Meanwhile, “zero-shot” is also used for pre-trained foundation- or diffusion-model approaches (Terris et al., 2025a; Spagnoletti et al., 2025; Chung et al., 2023) that handle unseen degradation tasks without specific fine-tuning, or adapt through self-supervised optimization from a small number of measurements. In contrast, our work considers a dataset-free, instance-level setting.

3. Theoretical Foundations of Compression-Based Inverse Solvers

3.1. Lossy compression

Let $\mathcal{Q} \subset \mathbb{R}^n$ be a signal class, and let (f, g) be a lossy compression code of rate R , i.e., $f : \mathcal{Q} \rightarrow \{1, \dots, 2^R\}$ and $g : \{1, \dots, 2^R\} \rightarrow \mathbb{R}^n$. We define the codebook $\mathcal{C} = \{g(i) : i = 1, \dots, 2^R\}$ and the signal-domain distortion as:

$$\delta = \sup_{\mathbf{x} \in \mathcal{Q}} \frac{1}{n} \|\mathbf{x} - g(f(\mathbf{x}))\|_2^2. \quad (4)$$

Distortion δ characterizes a reconstruction error guarantee over $\mathcal{Q}$ for the fixed codebook $\mathcal{C}$. This lossy compression model can serve as an implicit structure-aware prior, exploiting the fact that clean signals are more compressible.

3.2. Compression-based ML estimator

Consider a linear inverse problem that aims to reconstruct the clean signal $\mathbf{x} \in \mathbb{R}^n$ from an observation $\mathbf{y} \in \mathbb{R}^m$:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{z}, \mathbf{z} \sim \mathcal{N}(0, \sigma^2 I_m), \quad (5)$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a known linear degradation operator, and $\mathbf{z} \in \mathbb{R}^m$ represents additive Gaussian noise.

To analyze this, we consider a compression-based maximum-likelihood (ML) approach. Given an observation $\mathbf{y}$ and a lossy compression code (f, g) over $\mathcal{C}$ for the signal class $\mathcal{Q}$, the compression-based ML estimator is given as:

$$\hat{\mathbf{x}} = \arg \max_{\mathbf{c} \in \mathcal{C}} p(\mathbf{y} | \mathbf{c}) = \arg \min_{\mathbf{c} \in \mathcal{C}} \|\mathbf{y} - \mathbf{A}\mathbf{c}\|_2^2. \quad (6)$$

Derivations, along with other noise models, are provided in Appendix A.1. The proposed ML estimator leverages the rate constraint to restrict reconstructions to the codebook, thereby implicitly regularizing the solution toward structured signals. Under Gaussian noise, it reduces to a least-squares projection onto $\mathcal{C}$ in measurement domain. We characterize the induced measurement-domain distortion as:

$$\delta_A = \sup_{\mathbf{x} \in \mathcal{Q}} \frac{1}{n} \|\mathbf{A}(g(f(\mathbf{x})) - \mathbf{x})\|_2^2. \quad (7)$$

3.3. Theoretical foundations

We derive non-asymptotic error bounds for the proposed inverse solver, with denoising as a special case and signal-domain guarantees under mild assumptions.

Theorem 3.1 (Error bound for compression-based solver in general inverse problem). *Assume that $\mathbf{x} \in \mathcal{Q}$ and that (f, g) is a lossy compression code for $\mathcal{Q}$ with rate R and distortion δ . Consider the inverse problem $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{z}$, where $\mathbf{z} \sim \mathcal{N}(0, \sigma^2 I_m)$. Let $\hat{\mathbf{x}}$ denote the compression-based ML estimator, and let δ_A denote the measurement-domain distortion. Then, for any $\eta > 0$, the following holds with probability at least $1 - 2^{-\eta R}$:*

$$\frac{1}{\sqrt{n}} \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2 \leq \sqrt{\delta_A} + 2\sigma \sqrt{\frac{2 \ln 2(2 + \eta)R}{n}}. \quad (8)$$

In particular, if the operator norm satisfies $\|\mathbf{A}\|_{\text{op}} \leq L$:

$$\frac{1}{\sqrt{n}} \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2 \leq L\sqrt{\delta} + 2\sigma \sqrt{\frac{2 \ln 2(2 + \eta)R}{n}}. \quad (9)$$

Proof of Theorem 3.1 is provided in Appendix A.2. $\square$

Remark 3.2. The assumption $\|\mathbf{A}\|_{\text{op}} \leq L$ is mild and is only used to relate δ_A to δ , as it merely fixes the operator scale and can be enforced by appropriate normalization.

Theorem 3.1 provides a *measurement-domain fidelity guarantee* of the compression-based inverse solver. The bound consists of two components: (i) a distortion term that quantifies the approximation error induced by the compression codebook, and (ii) a rate-dependent noise term that scales with the noise level σ and the code rate R , capturing the statistical uncertainty induced by noise when the estimator is restricted to a finite-rate codebook. While stated for general linear inverse problems, the result specializes naturally to denoising by setting $\mathbf{A} = \mathbf{I}$, yielding a signal-domain error guarantee, as presented in Corollary 3.3.

Corollary 3.3 (Denoising specialization). *Consider the denoising setting $\mathbf{y} = \mathbf{x} + \mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$. Then $\delta_A = \delta$, and Theorem 3.1 directly yields, for any $\eta > 0$, with probability at least $1 - 2^{-\eta R}$,*

$$\frac{1}{\sqrt{n}} \|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \sqrt{\delta} + 2\sigma \sqrt{\frac{2 \ln 2(2 + \eta)R}{n}}. \quad (10)$$

Proof and analysis is provided in Appendix A.3. $\square$

Remark 3.4. In the denoising setting, our bound reduces to the same *two-term* structure, as existing analysis such as (Zafari et al., 2026), while arising as a direct specialization from a more general inverse-problem framework. A detailed comparison in Appendix A.3 shows that the relative tightness of two bounds depends on the operating regime, characterized by the confidence level η and compression rate R . In particular, our bound is more favorable in high-confidence or low-rate regimes, typical in zero-shot denoising to mitigate noise overfitting, while existing bounds (Zafari et al., 2026) can be tighter at large rates and low confidence regime.

To obtain an explicit signal-domain error bound, we impose an identifiability condition on the degradation operator over the codebook, leading to the following corollary.

Corollary 3.5 (Signal-domain error bound for compression-based solver in general inverse problem). *Assume the conditions of Theorem 3.1. Suppose that there exists $\alpha > 0$ such that $\|\mathbf{A}\mathbf{d}\|_2 \geq \alpha\|\mathbf{d}\|_2, \forall \mathbf{d} \in \mathcal{D}$, where $\mathcal{D} \triangleq \{\mathbf{c}_1 - \mathbf{c}_2 : \mathbf{c}_1, \mathbf{c}_2 \in \mathcal{C}\}$. Then, for any $\eta > 0$, the following holds with probability at least $1 - 2^{-\eta R}$:*

$$\frac{1}{\sqrt{n}} \|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \sqrt{\delta} + \frac{2}{\alpha} \sqrt{\delta_A} + \frac{2\sigma}{\alpha} \sqrt{\frac{2 \ln 2(2 + \eta)R}{n}}. \quad (11)$$

In particular, if $\|\mathbf{A}\|_{\text{op}} \leq L$, then $\delta_A \leq L^2 \delta$ and hence

$$\frac{1}{\sqrt{n}} \|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \left(1 + \frac{2L}{\alpha}\right) \sqrt{\delta} + \frac{2\sigma}{\alpha} \sqrt{\frac{2 \ln 2(2 + \eta)R}{n}}. \quad (12)$$

Proof of Corollary 3.5 are provided in Appendix A.4. $\square$

Remark 3.6. The condition $\|\mathbf{A}\mathbf{d}\|_2 \geq \alpha\|\mathbf{d}\|_2$ is an *injectivity* assumption of the forward operator restricted to the codebook difference set, which is standard in model-based inverse-problem analysis. It holds trivially for denoising ($\mathbf{A} = \mathbf{I}$). While $\mathbf{A}$ may be non-injective for tasks such as inpainting or super-resolution, learned compression codebooks often induce structured sets $\mathcal{D}$ on which this condition holds empirically (e.g., random-mask inpainting). The bound is thus most informative when $\mathbf{A}$ is well behaved on $\mathcal{D}$; otherwise, the proposed scheme can still be used as zero-shot denoisers within plug-and-play frameworks (Terris et al., 2024; Zhang et al., 2021).

Notably, the bound is non-asymptotic and does not require an optimal compression code, indicating that effective restoration is achievable with a well-calibrated RD trade-off. While the above analysis assumes a fixed compression codebook, in practice, such codebooks need not be pre-designed or learned from external data. Instead, they can be learned directly from the degraded observation via entropy regularization, which empirically biases the representation toward natural signal structure. This insight motivates our proposed *Lottery Prior*, which constructs a *family* of implicit codebooks from a single observation in a zero-shot manner.

4. Lottery Prior

We now introduce *Lottery Prior* (Fig. 4), a zero-shot inverse solver that enforces compression-based regularization via entropy control and randomized architectural priors, while maintaining measurement consistency. We next detail (i) entropy regularization, (ii) randomized subnetwork selection, and (iii) zero-shot optimization with prior ensembling.

4.1. Entropy regularization

Multi-scale latent representation. To encourage multi-scale self-similarity, we introduce a hierarchical learnable latent $\mathbf{Z}$ using an L -level pyramid structure:

$$\mathbf{Z} \triangleq \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_L\}, \text{ with } \mathbf{z}_k \in \mathbb{R}^{\frac{H}{2^{k-1}} \times \frac{W}{2^{k-1}}}, \quad (13)$$

where H and W are the image height and width. The resulting latent provides a compact representation of source structures across scales, a key property of natural images.

Auto-regressive probability model. To impose entropy constraints, we adopt a factorized auto-regressive model (ARM) $r_\psi(\cdot)$ to estimate the distribution of the quantized latent $\hat{\mathbf{Z}}$. Each element $\hat{z}_{i,j}$ (the j -th element of $\mathbf{z}_i$) is conditioned on its C spatial neighbors $\mathbf{c}_{i,j} \in \mathbb{Z}^C$ as $p_\psi(\hat{\mathbf{z}}) = \prod_{i,j} p_\psi(\hat{z}_{i,j} | \mathbf{c}_{i,j})$, where $p_\psi(\hat{\mathbf{z}})$ is modeled as:

$$p_\psi(\hat{z}_{i,j} | \mathbf{c}_{i,j}) = \int_{\hat{z}_{i,j}-0.5}^{\hat{z}_{i,j}+0.5} g(z) dz, \quad (14)$$

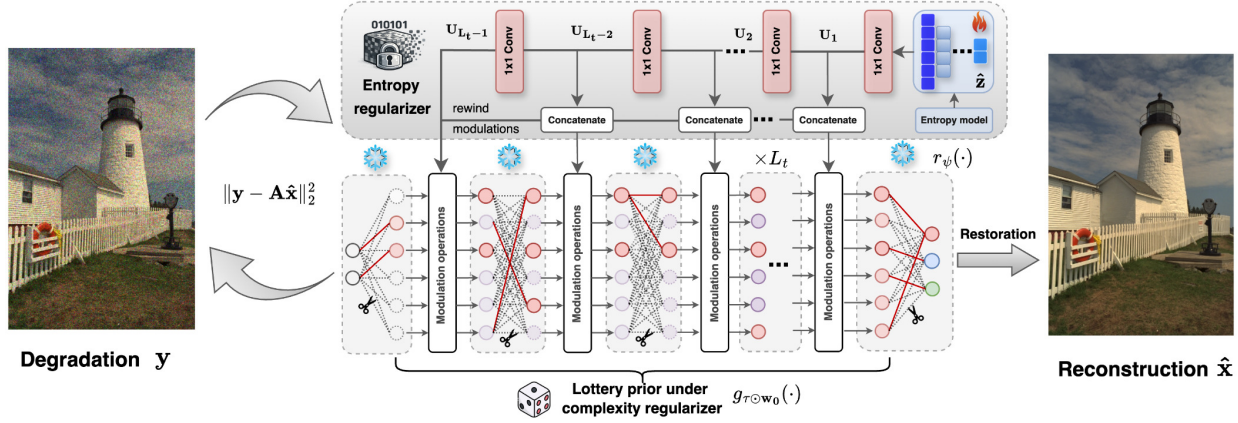


Figure 4. Architecture of Lottery Prior. A randomly initialized network maps coordinates to signal values by selecting subnetworks via a learned binary mask. Two implicit priors are enforced: an entropy prior, implemented through an entropy-regularized latent that generates layer-wise modulations, and an architectural prior, realized by subnetwork selection within a frozen random network. Solid red lines denote active weights, while dashed lines indicate masked connections.

where $g \sim \mathcal{L}(\mu_{i,j}, \sigma_{i,j})$ is a Laplace distribution with parameters $\mu_{i,j}, \sigma_{i,j} = r_\psi(\mathbf{c}_{i,j})$. The resulting rate term is:

$$R(\hat{\mathbf{z}}) = -\log_2 p_\psi(\hat{\mathbf{Z}}) = -\sum_{i,j} \log_2 p_\psi(\hat{z}_{i,j} | \mathbf{c}_{i,j}). \quad (15)$$

Beyond rate regularization, the ARM here also injects spatial context, which is particularly beneficial for inpainting, as later demonstrated in our experiments.

4.2. Randomized subnetwork selection

For an over-parameterized network $g_{\mathbf{W}_0}$, which maps pixel coordinates to signal values, we seek a high-performing subnetwork $g_{\tau \odot \mathbf{W}_0}$ via a learned mask τ , which serves as a prior by learning signal statistics into the network structure while remaining consistent with the degradation.

Subnetwork selection. The randomized network $g_{\tau \odot \mathbf{W}_0}$ consists of L_t masked linear layers with latent modulations, where Fourier initialization (Shi et al., 2024) is adopted to mitigate spectral bias while preserving stable variance. Subnetworks are selected via a learnable score matrix $\mathbf{P}$ with the same shape as $\mathbf{W}_0$, where each element scores the corresponding weight. The binary mask τ is obtained by activating the top $r_a\%$ of weights according to $\mathbf{P}$ while zeroing out the rest. Formally, the output of neuron k in layer i , denoted as $v_k^{(i)}$, is given by: $v_k^{(i)} = \sigma \left(\sum_{j=1}^{d_{i-1}} \tau_{k,j}^{(i-1)} w_{kj}^{(i-1)} m(v_j^{(i-1)}) \right)$, where $\tau_{k,j}$ is obtained by thresholding $\mathbf{P}$, $m(\cdot)$ denotes modulation function, σ the activation function, and $w_{kj}^{(i-1)}$ the weight connecting the k -th neuron in i -th layer to the j -th neuron of the $(i-1)$ -th layer. The scores in $\mathbf{P}$ are optimized using a straight-through estimator (Wortsman et al., 2019; Wu et al., 2025) under measurement-consistency constraints between $\mathbf{A}\hat{\mathbf{x}}$ and $\mathbf{y}$, together with entropy-based rate regularization.

Latent modulation. To simplify the learning process and improve restoration quality, we introduce a lightweight modulation network $f_\theta(\cdot)$ that maps entropy-constrained latents to layer-wise modulation signals. As shown in Fig. 4, the quantized $\hat{\mathbf{Z}}$ is upsampled via transposed convolutions, yielding $\mathbf{U}_0 = [\mathbf{u}_1; \mathbf{u}_2; \dots; \mathbf{u}_L] \in \mathbb{Z}^{L \times HW}$, which are processed by convolutional operations in f_θ to produce layer-wise modulations $\mathbf{U}_i$. Following the rewind strategy introduced in (Wu et al., 2025), modulation for i -th layer of $g_{\tau \odot \mathbf{W}_0}$ is constructed by reverse concatenation: $\mathbf{M}_i = \text{Concat}(\mathbf{U}_{L_t-1}, \mathbf{U}_{L_t-2}, \dots, \mathbf{U}_{L_t-i})$, and concatenated with the layer output $\mathbf{F}_i$ to produce $\mathbf{G}_i = m(\mathbf{F}_i) \triangleq \text{Concat}(\mathbf{M}_i, \mathbf{F}_i)$. Intuitively, this multi-scale modulation reinforces a DIP-style architectural prior by propagating structured information across layers, enabling effective subnetwork selection and improved restoration.

4.3. Zero-shot learning and prior ensemble

Optimization strategy. Lottery Prior is optimized in a zero-shot manner by learning the parameter set $\Omega \triangleq \{\mathbf{Z}, \psi, \theta, \tau\}$ from a single degraded observation $\mathbf{y}$. Compared with Eq. (3), additional parameters ψ and θ are introduced to facilitate the restoration process. The optimization minimizes the following objective:

$$\mathcal{L}(\Omega) = D(\mathbf{y}, \mathbf{A}\hat{\mathbf{x}}) + \lambda R(\hat{\mathbf{Z}}), \text{ with } \hat{\mathbf{x}} = g_{\tau \odot \mathbf{W}_0}(f_\theta(\hat{\mathbf{Z}})), \quad (16)$$

where $D(\cdot)$ is the mean-squared error enforcing measurement consistency, and λ is chosen empirically according to the degradation level. Training details are provided in Appendix B.4.

Restoration stage. To exploit prior diversity, we vary random seeds and mask ratios to obtain a family of reconstructions $\{\hat{\mathbf{x}}_1, \dots, \hat{\mathbf{x}}_k\}$, which are aggregated as: $\mathbf{x}^* =$

Table 1. Denoising performance comparison under AWGN and Poisson Noise, average PSNR(dB) and SSIM are reported. Best results are in **bold**, second-best are underlined. The number of effective network parameters reflects the size of the architectural prior and indicates restoration efficiency at inference.

Noise level σ or α	Method	Network parameters	AWGN, $\mathcal{N}(0, \sigma^2)$			Poisson, $\text{Poisson}(\alpha x)/\alpha$		
			Set11	Set13	Kodak24	Set11	Set13	Kodak24
15	JPEG2K	-	27.45 / 0.7699	26.69 / 0.7543	27.86 / 0.7457	22.35 / 0.5882	21.76 / 0.5494	22.56 / 0.5249
	BM3D	-	<u>32.22</u> / 0.8991	31.15 / 0.8808	32.37 / 0.8754	26.66 / 0.7505	25.64 / 0.6912	27.04 / 0.6900
	DIP	2.2M	29.11 / 0.7990	30.31 / 0.8570	31.42 / 0.8454	23.69 / 0.5863	25.14 / 0.6916	26.37 / 0.6761
	DD	0.1M	28.83 / 0.8215	29.22 / 0.8371	28.71 / 0.8016	24.37 / 0.6629	24.96 / 0.7006	25.59 / 0.6679
	S2S	1.0M	26.81 / 0.8158	20.61 / 0.6879	23.08 / 0.7695	21.75 / 0.6872	19.23 / 0.6553	22.52 / 0.7418
	ZS-N2S	0.2M	28.92 / 0.8495	18.18 / 0.5690	18.68 / 0.5540	25.06 / 0.7051	21.23 / 0.6066	22.24 / 0.6170
	ZS-N2N	22.3K	30.01 / 0.8169	30.95 / 0.8701	32.30 / 0.8650	24.04 / 0.5766	25.37 / 0.6878	26.80 / 0.6757
	ZS-NCD	0.4M	31.35 / 0.8580	<u>31.93</u> / 0.8983	<u>33.18</u> / 0.9026	25.65 / 0.7132	26.44 / 0.7434	<u>27.64</u> / 0.7432
	Lottery Prior	0.7-3.3K	32.30 / 0.8922	32.74 / 0.8963	34.29 / 0.9118	<u>26.16</u> / 0.7327	26.35 / 0.7702	28.23 / 0.7778
25	JPEG2K	-	24.91 / 0.6997	24.32 / 0.6676	25.43 / 0.6550	23.03 / 0.6108	22.65 / 0.5952	23.58 / 0.5680
	BM3D	-	29.79 / 0.8523	28.81 / 0.8213	29.98 / 0.8092	22.70 / 0.5741	22.17 / 0.5992	24.13 / 0.5931
	DIP	2.2M	26.60 / 0.7128	27.85 / 0.7837	28.90 / 0.7738	24.94 / 0.6512	26.13 / 0.7289	27.49 / 0.7243
	DD	0.1M	26.93 / 0.7530	27.40 / 0.7832	27.62 / 0.7496	25.48 / 0.7022	26.04 / 0.7373	26.56 / 0.7060
	S2S	1.0M	23.32 / 0.7306	17.95 / 0.5998	20.69 / 0.6949	23.40 / 0.7355	20.18 / 0.6927	23.09 / 0.7674
	ZS-N2S	0.2M	27.30 / 0.7971	20.39 / 0.6200	20.89 / 0.6156	26.01 / 0.7478	21.19 / 0.6312	21.47 / 0.6277
	ZS-N2N	22.3K	27.18 / 0.7173	28.36 / 0.8001	29.54 / 0.7798	25.40 / 0.6432	26.75 / 0.7455	28.21 / 0.7374
	ZS-NCD	0.4M	28.93 / 0.8079	<u>29.33</u> / 0.8351	<u>30.60</u> / 0.8144	<u>27.10</u> / 0.7431	<u>27.60</u> / 0.7827	<u>28.77</u> / 0.7677
	Lottery Prior	0.7-3.3K	<u>29.73</u> / 0.8420	30.00 / 0.8414	31.48 / 0.8580	27.64 / 0.7734	27.89 / 0.8088	29.55 / 0.8115

$\mathcal{G}(\{\hat{\mathbf{x}}_i\}_{i=1}^k)$. We use simple linear averaging for denoising and noisy super-resolution, which is sufficient for state-of-the-art performance, and report single-prior results for other inverse problems, such as image inpainting. Significant performance improvements are expected with more advanced aggregation strategies (Terris et al., 2025b).

5. Experimental results

We evaluated *Lottery Prior* on three canonical inverse-problems: (i) Denoising, including AWGN $\mathcal{N}(0, \sigma^2)$ with $\sigma \in \{15, 25\}$ and Poisson noise $\text{Poisson}(\alpha x)$ with $\alpha \in \{15, 25\}$; (ii) Noisy super-resolution, using $\times 4$ and $\times 8$ bicubic downsampling followed by additive Gaussian noise ($\sigma = 15$); and (iii) Inpainting, covering text removal, region completion and random pixel masking. Experimental setup and implementations are detailed in Appendix B.4.

5.1. Denoising under AWGN and Poisson Noise

We evaluate denoising performance under AWGN and Poisson noise on Set11 (Zhang et al., 2017), Set13 (Zeyde et al., 2010) (192 $\times$ 192 central crops), and Kodak24. Comparisons against representative zero-shot methods (see Appendix B for baseline details), including JPEG2K, BM3D, DIP, DD, ZS-N2S, S2S, ZS-N2N, and ZS-NCD, are reported in Table 1. Overall, *Lottery Prior* consistently achieves the best performance across all noise levels and datasets. Pixel-masking approaches such as ZS-N2S and S2S exhibit significant degradation on high-resolution images, likely due to blind-spot masking that limits detail recovery. Notably, *Lottery Prior* shows an increasing advantage over the previ-

Table 2. Super-resolution PSNR on Set5 with AWGN ($\sigma = 15$).

4 $\times$ super-resolution						
	Baby	Bird	Butterfly	Head	Woman	Avg.
No prior	21.56	20.59	17.25	21.47	19.64	20.10
Bicubic	25.04	24.41	19.89	24.63	22.64	23.32
BM3D+Bicubic	28.17	<u>25.56</u>	20.33	27.00	23.88	24.99
DIP	24.07	21.37	19.17	21.53	20.70	21.37
Ours	28.66	26.06	22.48	27.26	24.87	25.87
LapSRN	26.27	25.07	20.61	25.71	23.41	24.21
BM3D+LapSRN	<u>28.21</u>	25.51	<u>20.92</u>	<u>27.03</u>	<u>24.19</u>	<u>25.17</u>
8 $\times$ super-resolution						
	Baby	Bird	Butterfly	Head	Woman	Avg.
No prior	20.46	18.85	15.50	20.84	17.64	18.66
Bicubic	23.26	21.55	16.29	23.59	19.88	20.91
BM3D+Bicubic	24.81	21.68	16.32	24.99	20.30	21.62
DIP	21.01	20.15	16.55	21.00	19.01	19.55
Ours	25.46	<u>21.66</u>	17.28	25.06	20.90	22.07
LapSRN	24.25	21.51	16.18	24.32	20.26	21.30
BM3D+LapSRN	<u>25.20</u>	21.42	16.21	<u>24.98</u>	<u>20.49</u>	<u>21.66</u>

ous state-of-the-art ZS-NCD as image resolution increases, highlighting the benefit of our global structural prior over patch-wise neural compression. Moreover, our method is lightweight, avoids sensitive early stopping, and yields more stable restorations (see Appendix C.3). These results indicate strong potential for integrating lightweight implicit codecs into future plug-and-play zero-shot inverse solvers.

5.2. Super-Resolution with Noisy Observations

We report the performance of noisy super-resolution on Set5 in Table 2. More metrics, such as LPIPS and MS-SSIM,

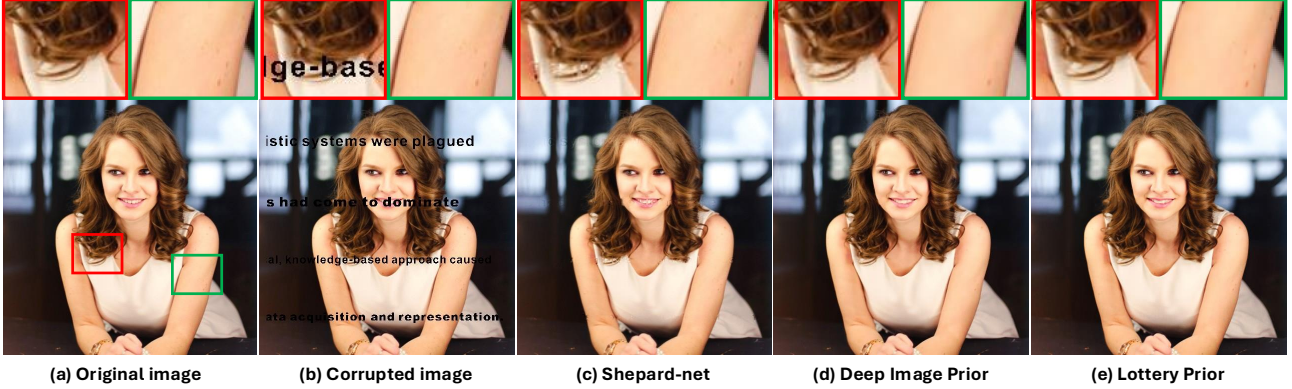


Figure 5. Compared with Shepard-net (Ren et al., 2015) and DIP, *Lottery Prior* preserves finer details without halo artifacts or early stopping, while DIP loses fidelity due to early stopping.

are reported in Table 11 of our appendix. DIP performs poorly in this challenging task due to its sensitivity to noise and reliance on early stopping, often requiring task-specific training strategies or architectural modifications, which limit its applicability to composite degradations. We additionally included a supervised super-resolution baseline LapSRN, combined with BM3D denoising. Despite operating in a zero-shot setting, *Lottery Prior* consistently achieves the highest PSNR across all test images. This demonstrates the robustness of compression-induced rate regularization. Qualitative comparisons in Figs. 8 and 9 further illustrate the superiority of *Lottery Prior*. In our experiments, the proposed method already demonstrates leading performance in all three metrics (PSNR, MS-SSIM, and LPIPS), and, in some cases, surpasses even strong supervised priors

5.3. Image Inpainting

Next, we evaluate the inpainting performance on three tasks: (i) *text removal*, (ii) *random mask inpainting*, and (iii) *region inpainting*. Text removal results in Fig. 5 show that *Lottery Prior* preserves finer details with fewer artifacts than both supervised methods and DIP, while requiring no early stopping and operating in a zero-shot manner. Random mask inpainting results are reported in Table 3 and Fig. 6, where our method demonstrates strong performance with accurate detail completion, benefiting from auto-regressive entropy modeling over nearby contextual cues. Additional visual results are provided in Fig. 10.

Region inpainting results in Figs. 11 and 12 further show that large missing areas can be plausibly completed using surrounding context, indicating that randomized masked networks act as an implicit structural prior similar to DIP. Fig. 12 illustrates the trade-off between rate constraints and inpainting fidelity. While generation capability is naturally limited by available context and model capacity, all results here use only a single prior; leveraging multiple lottery

Table 3. Random-mask inpainting PSNR on Set5 (mask ratio ρ).

	Baby	Bird	Butterfly	Head	Woman	Avg.
Deep image prior ($\rho = 0.8$)	29.56	29.93	23.87	26.33	25.99	27.14
Lottery prior ($\rho = 0.8$)	31.25	30.66	25.11	27.99	27.53	28.51
Deep image prior ($\rho = 0.5$)	31.87	35.38	28.31	29.66	31.13	31.27
Lottery prior ($\rho = 0.5$)	35.46	36.28	30.61	30.65	33.09	33.22

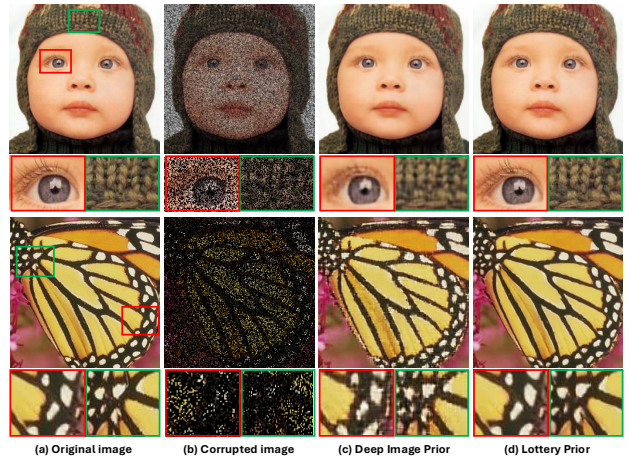


Figure 6. Visualization of inpainting results with mask ratios 0.5 (top) and 0.8 (bottom). *Lottery Prior* combines auto-regressive entropy modeling for spatial context with a global implicit prior from the randomized codec, enabling effective detail completion.

priors with stronger architectures and aggregation strategies is expected to yield further gains.

5.4. Task and modality adaptation

We further evaluated our method on JPEG artifact removal using AFHQ dataset (the first image from each class), with results reported in Table 5. The results show that our framework is not restricted to linear operators and consistently outperforms DIP. Although zero-shot JPEG artifact removal is challenging due to its nonlinear quantization degradation,

Table 4. Task adaptation experiments of RAM and Lottery Prior on demosaicing tasks, where the models are evaluated on classic samples (randomly selected from Set5/13 for testing; remaining samples used for RAM finetuning). RAM is pretrained on the LSDIR dataset (84,991 high-quality images) across 6 inverse tasks (such as deblur or inpainting) and then fine-tuned in a self-supervised manner (10 measurements), while Lottery Prior is trained using a single prior from a single degradation without external data.

Classic samples (randomly selected from Set5/13 for testing; remaining samples used for finetuning)						
PSNR	Baby	Butterfly	Woman	Coastguard	Foreman	Avg.
Lottery prior	33.17	31.15	32.77	31.59	32.43	32.22
RAM (Pretrained)	32.58	30.77	31.86	30.63	31.08	31.38
RAM (Finetuned N=10)	37.62	33.68	37.71	38.77	36.58	36.87

Table 5. Noisy JPEG artifact removal ($Q = 5, \sigma = 25$).

PSNR / MS-SSIM	Cat	Dog	Wild	Avg.
Deep image prior	25.17/0.81	24.09/0.77	23.13/0.77	24.13/0.79
Lottery prior	26.62/0.88	25.27/0.82	24.73/0.81	25.54/0.83

the proposed compression-based prior remains effective.

We also compare with RAM (Terris et al., 2025a) on unseen tasks and modalities, including demosaicing (Table 4) and microscopy imaging (Table 12). RAM achieves strong results, especially after self-supervised fine-tuning, while our method performs consistently without pretraining. Since RAM relies on extensive supervised pretraining, we report these results only to calibrate our dataset-free method against pretrained foundation-model priors. Additional MRI experiments are provided in Table 13 of the Appendix, further demonstrating task and modality adaptation.

Note that highly ill-conditioned degradations, such as JPEG artifact removal and MRI reconstruction, remain challenging in the pure zero-shot setting. While our method shows initial effectiveness, fully recovering signals using only compressibility and untrained priors remains difficult. Promising directions include approximate forward models, perceptual coding objectives, and iterative or hybrid schemes to better balance data consistency and prior regularization.

5.5. Ablation study and complexity analysis

Detailed ablation studies for each component are reported in Tables 8, 9, and 10, validating the proposed design and showing that a moderate model size is preferred in practice. We further note that, under the randomized network setting, different architectures can induce diverse random codec priors; judiciously combining such priors may further improve performance, which we leave for future work.

We report runtime and computational complexity in Table 6 using the first three Set11 images, with backward MACs approximated as twice the forward MACs (Baydin et al., 2018). Compared to ZS-NCD (0.4M parameters, officially reported 40 min/image) and lightweight DIP variants (1-2 min/image; 2.2M parameters for the original model), our

Table 6. Runtime (sec/1k steps) and complexity (MACs/pixel) for zero-shot methods to denoise a Set11 image with $\sigma = 15$ and mask ratio 0.5, on NVIDIA 3090Ti GPU and AMD 5900X CPU.

Methods	Run time	Required steps	Per-step complexity
ZS-NCD	96.91	20k	2350.08k
DIP	22.38	2k-4k	899.78k
Lottery prior	12.38	5k-10k	10.37k

method is more efficient due to its compact design (0.7-3.3k parameters) and avoidance of patch-wise redundancy, yielding lower per-iteration cost and stronger performance. Step-wise results in Table 14 further show a flexible time-performance trade-off. Our framework also supports flexible prior aggregation: a single prior (5-10k steps, ~ 2 min) suffices for structured tasks such as inpainting, while aggregating a few priors (e.g., 5 mask ratios, ~ 5 -10 min) further improves denoising.

6. Conclusion and future work

We presented *Lottery Prior*, a lightweight zero-shot framework for inverse problems that combines entropy-based compression regularization with randomized architectural priors. By constraining the solution space via rate control and leveraging the structural bias of random networks, it induces a family of implicit priors from a single observation without external training. We provide non-asymptotic error bounds for compression-based solvers, showing how RD constraint acts as an implicit regularizer with provable guarantees. Extensive experiments demonstrate state-of-the-art results on denoising, noisy super-resolution, and inpainting with significantly fewer effective parameters.

Limitations and future work. While *Lottery Prior* excels on composite degradations, its gains are less pronounced in the case of severely ill-conditioned forward operators. From an information-theoretic perspective, these settings are dominated by operator-induced ambiguity rather than signal uncertainty, reducing the effectiveness of entropy-based regularization. A promising direction for future work is to integrate *Lottery Prior* into plug-and-play frameworks that explicitly handle the forward operator.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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Appendix

Appendix Contents

- A. Proof of theorem and corollary
- B. Implementation details
- C. Visualization
- D. Ablation studies and more experiments
- E. Detailed datapoints

A. Proof of theorem and corollary

A.1. Derivations of ML estimator

Recall of [The ML estimator and noise model]: In this paper, we consider the standard Gaussian linear inverse problem:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{z}, \mathbf{z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_m). \quad (17)$$

Given a noisy observation $\mathbf{y}$ and a lossy compression code (f, g) with codebook $\mathcal{C}$ for the signal class $\mathcal{Q}$, the compression-based ML estimator is given as:

$$\hat{\mathbf{x}} = \arg \max_{\mathbf{c} \in \mathcal{C}} p(\mathbf{y} \mid \mathbf{c}) = \arg \min_{\mathbf{c} \in \mathcal{C}} \|\mathbf{y} - \mathbf{A}\mathbf{c}\|_2^2. \quad (18)$$

We emphasize that the derived ML estimator is specific to Gaussian noise.

Proof. Under the additive Gaussian noise model $\mathbf{z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_m)$, the conditional distribution of the observation $\mathbf{y}$ given a candidate signal $\mathbf{c} \in \mathcal{C}$ is:

$$p(\mathbf{y} \mid \mathbf{c}) = \frac{1}{(2\pi\sigma^2)^{m/2}} \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{A}\mathbf{c}\|_2^2\right). \quad (19)$$

Taking the logarithm yields the log-likelihood:

$$\log p(\mathbf{y} \mid \mathbf{c}) = -\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{A}\mathbf{c}\|_2^2 - \frac{m}{2} \log(2\pi\sigma^2). \quad (20)$$

Since the second term is independent of $\mathbf{c}$, maximizing the log-likelihood over $\mathbf{c} \in \mathcal{C}$ is equivalent to minimizing the squared Euclidean distance:

$$\arg \max_{\mathbf{c} \in \mathcal{C}} p(\mathbf{y} \mid \mathbf{c}) = \arg \min_{\mathbf{c} \in \mathcal{C}} \|\mathbf{y} - \mathbf{A}\mathbf{c}\|_2^2. \quad (21)$$

□

Thus, under Gaussian noise, the compression-based ML estimator reduces to a least-squares projection onto the codebook by evaluating the likelihood of the observation under the conditional model associated with each candidate signal.

Remark A.1 (Other noise models). For noise models other than Gaussian, the maximum-likelihood (ML) estimator generally takes a different form. For instance, under Poisson noise with observations $y_i \sim \text{Poisson}(\alpha(\mathbf{A}\mathbf{x})_i)$ (each element of the measured signal), the corresponding ML estimator minimizes the Poisson negative log-likelihood:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{c} \in \mathcal{C}} \sum_i [\alpha(\mathbf{A}\mathbf{c})_i - y_i \log(\mathbf{A}\mathbf{c})_i]. \quad (22)$$

Theoretical guarantees for such noise models can be derived using arguments analogous to the Gaussian case, relying on appropriate inequalities. For clarity and interpretability, we therefore focus our theoretical guarantees on linear inverse problems with Gaussian noise in this work.

A.2. Proof of Theorem 3.1

Recall of Theorem 3.1 [Compression-based ML for General Inverse Problem]: Assume that $\mathbf{x} \in \mathcal{Q}$ and that (f, g) is a lossy compression code for $\mathcal{Q}$ with rate R and distortion δ . Consider the inverse problem $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{z}$, where $\mathbf{z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_m)$. Let $\hat{\mathbf{x}}$ denote the compression-based ML estimator, and δ_A denote the measurement-domain distortion. Then, for any $\eta > 0$, the following holds with probability at least $1 - 2^{-\eta R}$:

$$\frac{1}{\sqrt{n}} \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2 \leq \sqrt{\delta_A} + 2\sigma \sqrt{\frac{2 \ln 2(2 + \eta)R}{n}}. \quad (23)$$

In particular, if the operator norm of $\mathbf{A}$ satisfies $\|\mathbf{A}\|_{\text{op}} \leq L$, then

$$\frac{1}{\sqrt{n}} \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2 \leq L\sqrt{\delta} + 2\sigma \sqrt{\frac{2 \ln 2(2 + \eta)R}{n}}. \quad (24)$$

Proof of Theorem 3.1. Let $\mathbf{x}_c = g(f(\mathbf{x})) \in \mathcal{C}$. By definition of $\hat{\mathbf{x}}$,

$$\|\mathbf{y} - \mathbf{A}\hat{\mathbf{x}}\|_2^2 \leq \|\mathbf{y} - \mathbf{A}\mathbf{x}_c\|_2^2. \quad (25)$$

Substituting $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{z}$ into two sides of Eqn. (25) yields

$$\|\mathbf{A}\mathbf{x} + \mathbf{z} - \mathbf{A}\hat{\mathbf{x}}\|_2^2 \leq \|\mathbf{A}\mathbf{x} + \mathbf{z} - \mathbf{A}\mathbf{x}_c\|_2^2, \quad (26)$$

i.e.,

$$\|\mathbf{z} - \mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2^2 \leq \|\mathbf{z} + \mathbf{A}(\mathbf{x} - \mathbf{x}_c)\|_2^2. \quad (27)$$

Expanding both sides leads to:

$$\|\mathbf{z} - \mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2^2 = \|\mathbf{z}\|_2^2 - 2\mathbf{z}^\top \mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}) + \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2^2, \quad (28)$$

and

$$\|\mathbf{z} + \mathbf{A}(\mathbf{x} - \mathbf{x}_c)\|_2^2 = \|\mathbf{z}\|_2^2 + 2\mathbf{z}^\top \mathbf{A}(\mathbf{x} - \mathbf{x}_c) + \|\mathbf{A}(\mathbf{x} - \mathbf{x}_c)\|_2^2. \quad (29)$$

Removing $\|\mathbf{z}\|_2^2$ and rearranging yields:

$$\|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2^2 \leq \|\mathbf{A}(\mathbf{x} - \mathbf{x}_c)\|_2^2 + 2\mathbf{z}^\top \mathbf{A}((\hat{\mathbf{x}} - \mathbf{x}) + (\mathbf{x} - \mathbf{x}_c)) = \|\mathbf{A}(\mathbf{x}_c - \mathbf{x})\|_2^2 + 2\mathbf{z}^\top \mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}_c). \quad (30)$$

We therefore have:

$$\|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2^2 \leq \|\mathbf{A}(\mathbf{x}_c - \mathbf{x})\|_2^2 + 2\mathbf{z}^\top \mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}_c). \quad (31)$$

To allow for a tractable probabilistic analysis, we then aim to upper bound it by a supremum over a finite set. Recall that there exist indices $i, j \in \{1, \dots, 2^R\}$ such that $\hat{\mathbf{x}} = g(i)$ and $\mathbf{x}_c = g(j)$, and therefore $\hat{\mathbf{x}} - \mathbf{x}_c = g(i) - g(j) \in \mathcal{D}$, where $\mathcal{D} \triangleq \{\mathbf{c}_i - \mathbf{c}_j : \mathbf{c}_i, \mathbf{c}_j \in \mathcal{C}\}$ for any i, j . Note that since $\mathcal{C}$ is a finite codebook with $|\mathcal{C}| = 2^R$, the difference set $\mathcal{D}$ is finite with $|\mathcal{D}| \leq 2^{2R}$. Consequently, for the particular random vector $\hat{\mathbf{x}} - \mathbf{x}_c$, we have:

$$\mathbf{z}^\top \mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}_c) \leq \sup_{\mathbf{d} \in \mathcal{D}} \mathbf{z}^\top \mathbf{A}\mathbf{d}. \quad (32)$$

Given an arbitrary $\mathbf{d} \in \mathbb{R}^n$, since $\mathbf{z} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_m)$ and $\mathbf{A}\mathbf{d} \in \mathbb{R}^m$ is deterministic when $\mathbf{d}$ is fixed, the random variable $\mathbf{z}^\top \mathbf{A}\mathbf{d}$ can be given as:

$$\mathbf{z}^\top \mathbf{A}\mathbf{d} = \langle \mathbf{z}, \mathbf{A}\mathbf{d} \rangle = (\mathbf{A}\mathbf{d})^\top \mathbf{z}, \quad (33)$$

which is a Gaussian with the corresponding mean and variance as:

$$\mathbb{E}[\mathbf{z}^\top \mathbf{A}\mathbf{d}] = (\mathbf{A}\mathbf{d})^\top \mathbb{E}[\mathbf{z}] = 0, \quad (34)$$

$$\text{Var}(\mathbf{z}^\top \mathbf{A} \mathbf{d}) = \mathbb{E}[(\mathbf{z}^\top \mathbf{A} \mathbf{d})^2] = \mathbb{E}[(\mathbf{A} \mathbf{d})^\top \mathbf{z} \mathbf{z}^\top (\mathbf{A} \mathbf{d})] = (\mathbf{A} \mathbf{d})^\top \mathbb{E}[\mathbf{z} \mathbf{z}^\top] (\mathbf{A} \mathbf{d}). \quad (35)$$

Since $\mathbb{E}[\mathbf{z} \mathbf{z}^\top] = \sigma^2 \mathbf{I}_m$, it can be simplified as:

$$\text{Var}(\mathbf{z}^\top \mathbf{A} \mathbf{d}) = (\mathbf{A} \mathbf{d})^\top (\sigma^2 \mathbf{I}_m) (\mathbf{A} \mathbf{d}) = \sigma^2 \|\mathbf{A} \mathbf{d}\|_2^2. \quad (36)$$

Therefore, for any fixed $\mathbf{d}$, we have

$$X_d \triangleq \mathbf{z}^\top \mathbf{A} \mathbf{d} \sim \mathcal{N}(0, \sigma^2 \|\mathbf{A} \mathbf{d}\|_2^2). \quad (37)$$

Let $U_d \triangleq X_d / (\sigma \|\mathbf{A} \mathbf{d}\|_2) \sim \mathcal{N}(0, 1)$. Using the standard Gaussian tail bound $\Pr(U_d \geq t) \leq e^{-t^2/2}$ for any $t > 0$, we obtain:

$$\Pr(X_d \geq \sigma \|\mathbf{A} \mathbf{d}\|_2 t) = \Pr(U_d \geq t) \leq e^{-t^2/2}. \quad (38)$$

Applying a union bound over the finite set $\mathcal{D}$ yields

$$\Pr(\exists \mathbf{d} \in \mathcal{D} : X_d \geq \sigma \|\mathbf{A} \mathbf{d}\|_2 t) \leq \sum_{\mathbf{d} \in \mathcal{D}} e^{-t^2/2} = |\mathcal{D}| e^{-t^2/2}. \quad (39)$$

Equivalently, with probability at least $1 - |\mathcal{D}| e^{-t^2/2}$, we obtain that for all $\mathbf{d} \in \mathcal{D}$ we satisfy $X_d \leq \sigma \|\mathbf{A} \mathbf{d}\|_2 t$.

Since $|\mathcal{C}| = 2^R$, we have $|\mathcal{D}| \leq |\mathcal{C}|^2 = 2^{2R}$. Hence,

$$\Pr(\exists \mathbf{d} \in \mathcal{D} : \mathbf{z}^\top \mathbf{A} \mathbf{d} \geq \sigma \|\mathbf{A} \mathbf{d}\|_2 t) \leq 2^{2R} e^{-t^2/2}. \quad (40)$$

We next choose a t such that the right-hand side is upper bounded by a form of $2^{-\eta R}$, i.e.,

$$2^{2R} e^{-t^2/2} \leq 2^{-\eta R}. \quad (41)$$

Taking logarithms on both sides yields

$$\frac{t^2}{2} \geq (2 + \eta) R \ln 2, \quad (42)$$

and therefore we set

$$t \triangleq \sqrt{2(2 + \eta) R \ln 2}. \quad (43)$$

With this choice, it follows that with probability at least $1 - 2^{-\eta R}$, the following inequality holds for all $\mathbf{d} \in \mathcal{D}$:

$$\mathbf{z}^\top \mathbf{A} \mathbf{d} \leq \sigma \|\mathbf{A} \mathbf{d}\|_2 t. \quad (44)$$

That is, with probability at least $1 - 2^{-\eta R}$ ¹:

$$\mathbf{z}^\top \mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}_c) \leq \sigma \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}_c)\|_2 t, \quad (45)$$

where $t \triangleq \sqrt{2(2 + \eta) R \ln 2}$.

Substituting this bound into (31) yields:

$$\|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2^2 \leq \|\mathbf{A}(\mathbf{x}_c - \mathbf{x})\|_2^2 + 2\sigma t \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}_c)\|_2. \quad (46)$$

To further simplify, let $\mathbf{e} \triangleq \hat{\mathbf{x}} - \mathbf{x}$ and $\mathbf{e}_c \triangleq \mathbf{x}_c - \mathbf{x}$. Since $\hat{\mathbf{x}} - \mathbf{x}_c = \mathbf{e} - \mathbf{e}_c$, by the triangle inequality, we have

$$\|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}_c)\|_2 \leq \|\mathbf{A} \mathbf{e}\|_2 + \|\mathbf{A} \mathbf{e}_c\|_2. \quad (47)$$

¹The probability guarantee differs from previous work focusing on denoising (Zafari et al., 2026) only in constant factors. Specifically, the additional constant factor appearing in expressions such as $2^{-\eta R+2}$ in previous work (Zafari et al., 2026) arises from the use of a union bound over two events together with a two-sided Gaussian tail bound. In contrast, our analysis employs a one-sided tail bound tailored to the more general linear inverse problem setting. This choice avoids unnecessary constant slack in the probability estimate while preserving the same error scaling and recovering the denoising result as a special case.

By replacing Eqn. (47) into Eqn. (46), we have:

$$\|\mathbf{A}\mathbf{e}\|_2^2 \leq \|\mathbf{A}\mathbf{e}_c\|_2^2 + 2\sigma t(\|\mathbf{A}\mathbf{e}\|_2 + \|\mathbf{A}\mathbf{e}_c\|_2). \quad (48)$$

Eqn. (48) can be rearranged and written as:

$$(\|\mathbf{A}\mathbf{e}\|_2 - \sigma t)^2 \leq (\|\mathbf{A}\mathbf{e}_c\|_2 + \sigma t)^2, \quad (49)$$

which implies

$$\|\mathbf{A}\mathbf{e}\|_2 \leq \|\mathbf{A}\mathbf{e}_c\|_2 + 2\sigma t. \quad (50)$$

Thus, we obtain the following measurement-domain error bound:

$$\begin{aligned} \frac{1}{\sqrt{n}} \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2 &\leq \frac{1}{\sqrt{n}} \|\mathbf{A}(\mathbf{x}_c - \mathbf{x})\|_2 + 2\sigma \sqrt{\frac{2 \ln 2 (2 + \eta) R}{n}} \\ &\leq \sqrt{\delta_A} + 2\sigma \sqrt{\frac{2 \ln 2 (2 + \eta) R}{n}} \end{aligned} \quad (51)$$

Finally, if $\|\mathbf{A}\|_{\text{op}} \leq L$, recalling that $\|\mathbf{x} - \mathbf{x}_c\|_2^2 \leq n\delta$, it follows that $\|\mathbf{A}\mathbf{e}_c\|_2 \leq L\sqrt{n\delta}$. Consequently, we obtain

$$\frac{1}{\sqrt{n}} \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2 \leq L\sqrt{\delta} + 2\sigma \sqrt{\frac{2 \ln 2 (2 + \eta) R}{n}}. \quad (52)$$

□

Remark A.2 (One-sided vs. two-sided Gaussian tail bounds). In previous analysis over denoising task (Zafari et al., 2026), the proof employs a two-sided Gaussian tail bound over the form $\langle \mathbf{z}, \mathbf{d} / \|\mathbf{d}\|_2 \rangle \sim \mathcal{N}(0, \sigma^2)$, leading to probability bounds of the type $\Pr(|\langle \mathbf{z}, \mathbf{d} / \|\mathbf{d}\|_2 \rangle| \geq t) \leq 2 \exp(-t^2 / (2\sigma^2))$, where the inequalities involving absolute values arise from intermediate steps Eqn. (27) to (28). Notably, here two union-bound operations are operated.

In contrast, our analysis focuses on $\Pr(\langle \mathbf{z}, \mathbf{A}\mathbf{d} / \|\mathbf{A}\mathbf{d}\|_2 \rangle \geq t) \leq \exp(-t^2 / (2\sigma^2))$, using a one-sided Gaussian tail bound $\mathbb{P}(\langle \mathbf{z}, \mathbf{u} \rangle \geq t) \leq \exp(-t^2 / (2\sigma^2))$, with one union-bound operation introduced. This makes a difference of factor 2 before the exponentiation operations.

For intuition, we compare the true codeword $\mathbf{x}$ with a candidate $\mathbf{c} \in \mathcal{C}$. A decoding error occurs only if: $\|\mathbf{y} - \mathbf{A}\mathbf{c}\|_2^2 \leq \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2$. Substituting $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{z}$ and simplifying yields the equivalent condition $2\langle \mathbf{z}, \mathbf{A}(\mathbf{c} - \mathbf{x}) \rangle \geq \|\mathbf{A}(\mathbf{c} - \mathbf{x})\|_2^2$. Letting $\mathbf{u} = \mathbf{A}(\mathbf{c} - \mathbf{x})$, the error event becomes $\langle \mathbf{z}, \mathbf{u} \rangle \geq \frac{1}{2}\|\mathbf{u}\|_2^2$. Since $\langle \mathbf{z}, \mathbf{u} \rangle \sim \mathcal{N}(0, \sigma^2\|\mathbf{u}\|_2^2)$ (see Eqn. (33)), a decoding error requires an unusually large positive noise deviation along $\mathbf{u}$, so only one-sided Gaussian tail bound suffices.

A.3. Proof and analysis of Corollary 3.3

Proof of Corollary 3.3. Setting $\mathbf{A} = \mathbf{I}_n$ in Theorem 3.1 gives $\|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2 = \|\hat{\mathbf{x}} - \mathbf{x}\|_2$. Moreover, $\delta_A = \sup_{\mathbf{x} \in \mathcal{Q}} \frac{1}{n} \|\mathbf{A}(g(f(\mathbf{x})) - \mathbf{x})\|_2^2 = \sup_{\mathbf{x} \in \mathcal{Q}} \frac{1}{n} \|g(f(\mathbf{x})) - \mathbf{x}\|_2^2 = \delta$. Substituting these into the theorem yields the conclusion. □

We compare the denoising specialization of our measurement-domain bound with the corresponding result in previous zero-shot compression-based denoising literature (Zafari et al., 2026).

Our bound. In the denoising setting $\mathbf{A} = \mathbf{I}_n$ (hence $\delta_A = \delta$), for any $\eta > 0$, with probability at least $1 - 2^{-\eta R}$,

$$\frac{1}{\sqrt{n}} \|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \sqrt{\delta} + 2\sigma \sqrt{\frac{2 \ln 2 (2 + \eta) R}{n}}. \quad (53)$$

Previous result. The denoising bound reported in (Zafari et al., 2026) can be written as: for any $\tilde{\eta} > 0$, with probability at least $1 - 2^{-\tilde{\eta}R+2}$,

$$\frac{1}{\sqrt{n}} \|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \sqrt{\delta} + 2\sigma \sqrt{\frac{2 \ln 2 R}{n}} (1 + 2\sqrt{\tilde{\eta}}). \quad (54)$$

To compare the two bounds under the *same* scenario, we parameterize both results by a common $\eta > 0$ via

$$\tilde{\eta} = \eta + \frac{2}{R}. \quad (55)$$

Then, we can have $1 - 2^{-\tilde{\eta}R+2} = 1 - 2^{-(\eta+2/R)R+2} = 1 - 2^{-\eta R}$, so both (53) and (54) hold with probability at least $1 - 2^{-\eta R}$ after substituting (55) into (54).

After unifying the probability, both bounds share the same distortion term $\sqrt{\delta}$ and the same scaling $\sigma \sqrt{R/n}$, and differ only in the confidence-dependent constants. Define

$$C_{\text{ours}}(\eta) = \sqrt{2 + \eta}, \quad C_{\text{pre}}(\eta, R) = 1 + 2\sqrt{\eta + \frac{2}{R}}. \quad (56)$$

Under the same confidence level $1 - 2^{-\eta R}$, our bound is tighter if and only if

$$C_{\text{ours}}(\eta) \leq C_{\text{pre}}(\eta, R) \iff \sqrt{2 + \eta} \leq 1 + 2\sqrt{\eta + \frac{2}{R}}. \quad (57)$$

Regime analysis. Since $C_{\text{pre}}(\eta, R)$ is minimized as $R \rightarrow \infty$, the most stringent condition is obtained in this limit, yielding

$$\sqrt{2 + \eta} \leq 1 + 2\sqrt{\eta}. \quad (58)$$

This inequality holds if and only if

$$\eta \geq \eta^* \triangleq \frac{11 - 4\sqrt{7}}{9} \approx 0.0463. \quad (59)$$

Consequently, the following regimes can be identified:

- **High-confidence regime.** If $\eta \geq \eta^*$, then $C_{\text{ours}}(\eta) \leq C_{\text{pre}}(\eta, R)$ for all $R > 0$, and our bound is tighter.
- **Low-confidence, small-rate regime.** If $\eta < \eta^*$, then (57) holds provided that

$$R \leq \frac{8}{3(1 - \eta) - 2\sqrt{2 + \eta}}, \quad (60)$$

and our bound remains tighter.

- **Low-confidence, large-rate regime.** Only if $\eta < \eta^*$ and

$$R > \frac{8}{3(1 - \eta) - 2\sqrt{2 + \eta}}, \quad (61)$$

then $C_{\text{ours}}(\eta) > C_{\text{pre}}(\eta, R)$, and the bound in (Zafari et al., 2026) becomes tighter.

Summary. Our theoretical results provide a more general framework for a compression-based solver, which naturally specializes to the denoising setting and recovers the error bound of the same two-term structure as prior results, consisting of a distortion term and a rate-dependent term. Their relative tightness depends on the confidence parameter η and the compression rate R . In particular, our bound is tighter under high-confidence requirements or small compression rates, which are typical operating regimes in zero-shot denoising to mitigate noise overfitting, whereas for large rates and low-confidence requirements the bound in (Zafari et al., 2026) is tighter.

A.4. Proof of Corollary 3.5

Recall of Corollary 3.5 [Signal-domain error bound for compression-based solver in linear inverse problem]: Assume the conditions of Theorem 3.1. In addition, suppose that there exists $\alpha > 0$ such that

$$\|\mathbf{A}\mathbf{d}\|_2 \geq \alpha\|\mathbf{d}\|_2, \quad \forall \mathbf{d} \in \mathcal{D}, \quad (62)$$

where $\mathcal{D} \triangleq \{c_1 - c_2 : c_1, c_2 \in \mathcal{C}\}$. Then, for any $\eta > 0$, with probability at least $1 - 2^{-\eta R}$, it holds:

$$\frac{1}{\sqrt{n}}\|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \sqrt{\delta} + \frac{2}{\alpha}\sqrt{\delta_A} + \frac{2\sigma}{\alpha}\sqrt{\frac{2\ln 2(2+\eta)R}{n}}. \quad (63)$$

In particular, if $\|\mathbf{A}\|_{\text{op}} \leq L$, then $\delta_A \leq L^2\delta$ and hence

$$\frac{1}{\sqrt{n}}\|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \left(1 + \frac{2L}{\alpha}\right)\sqrt{\delta} + \frac{2\sigma}{\alpha}\sqrt{\frac{2\ln 2(2+\eta)R}{n}}. \quad (64)$$

Proof of Corollary 3.5. Recall that $\mathbf{x}_c = g(f(\mathbf{x})) \in \mathcal{C}$ denotes the reconstruction of $\mathbf{x}$ by the compression code. By the triangle inequality,

$$\|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \|\hat{\mathbf{x}} - \mathbf{x}_c\|_2 + \|\mathbf{x}_c - \mathbf{x}\|_2. \quad (65)$$

Since $\mathbf{x}_c \in \mathcal{C}$ and $\hat{\mathbf{x}} \in \mathcal{C}$, their difference belongs to the codeword-difference set $\mathcal{D}$. Thus, by the restricted injectivity assumption,

$$\|\hat{\mathbf{x}} - \mathbf{x}_c\|_2 \leq \frac{1}{\alpha}\|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}_c)\|_2. \quad (66)$$

Moreover, from triangle inequality, we have

$$\|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x}_c)\|_2 \leq \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2 + \|\mathbf{A}(\mathbf{x} - \mathbf{x}_c)\|_2. \quad (67)$$

Combining (65)–(67) yields:

$$\|\hat{\mathbf{x}} - \mathbf{x}\|_2 \leq \frac{1}{\alpha}\|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2 + \frac{1}{\alpha}\|\mathbf{A}(\mathbf{x} - \mathbf{x}_c)\|_2 + \|\mathbf{x}_c - \mathbf{x}\|_2. \quad (68)$$

Now, based on Theorem 3.1 (Eqn. (50)), we have:

$$\frac{1}{\sqrt{n}}\|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|_2 \leq \frac{1}{\sqrt{n}}\|\mathbf{A}(\mathbf{x}_c - \mathbf{x})\|_2 + 2\sigma\sqrt{\frac{2\ln 2(2+\eta)R}{n}}. \quad (69)$$

Substituting (69) into (68) and dividing by $\sqrt{n}$, we obtain

$$\begin{aligned} \frac{1}{\sqrt{n}}\|\hat{\mathbf{x}} - \mathbf{x}\|_2 &\leq \frac{1}{\alpha} \left[\frac{1}{\sqrt{n}}\|\mathbf{A}(\mathbf{x}_c - \mathbf{x})\|_2 + 2\sigma\sqrt{\frac{2\ln 2(2+\eta)R}{n}} \right] + \frac{1}{\alpha} \frac{1}{\sqrt{n}}\|\mathbf{A}(\mathbf{x} - \mathbf{x}_c)\|_2 + \frac{1}{\sqrt{n}}\|\mathbf{x}_c - \mathbf{x}\|_2 \\ &= \frac{2}{\alpha} \cdot \frac{1}{\sqrt{n}}\|\mathbf{A}(\mathbf{x}_c - \mathbf{x})\|_2 + \frac{2\sigma}{\alpha}\sqrt{\frac{2\ln 2(2+\eta)R}{n}} + \frac{1}{\sqrt{n}}\|\mathbf{x}_c - \mathbf{x}\|_2. \end{aligned} \quad (70)$$

Finally, using the signal-domain distortion definition,

$$\frac{1}{\sqrt{n}}\|\mathbf{x}_c - \mathbf{x}\|_2 \leq \sqrt{\delta}, \quad (71)$$

we can have:

$$\begin{aligned} \frac{1}{\sqrt{n}}\|\hat{\mathbf{x}} - \mathbf{x}\|_2 &\leq \frac{2}{\alpha} \cdot \frac{1}{\sqrt{n}}\|\mathbf{A}(\mathbf{x}_c - \mathbf{x})\|_2 + \frac{2\sigma}{\alpha}\sqrt{\frac{2\ln 2(2+\eta)R}{n}} + \sqrt{\delta} \\ &= \frac{2}{\alpha}\sqrt{\delta_A} + \frac{2\sigma}{\alpha}\sqrt{\frac{2\ln 2(2+\eta)R}{n}} + \sqrt{\delta} \end{aligned} \quad (72)$$

Considering that the operator-norm bound $\|\mathbf{A}\|_{\text{op}} \leq L$ gives

$$\frac{1}{\sqrt{n}} \|\mathbf{A}(\mathbf{x}_c - \mathbf{x})\|_2 \leq \|\mathbf{A}\|_{\text{op}} \cdot \frac{1}{\sqrt{n}} \|\mathbf{x}_c - \mathbf{x}\|_2 \leq L\sqrt{\delta}, \quad (73)$$

we obtain:

$$\begin{aligned} \frac{1}{\sqrt{n}} \|\hat{\mathbf{x}} - \mathbf{x}\|_2 &\leq \frac{2}{\alpha} \cdot \frac{1}{\sqrt{n}} \|\mathbf{A}(\mathbf{x}_c - \mathbf{x})\|_2 + \frac{2\sigma}{\alpha} \sqrt{\frac{2 \ln 2(2 + \eta)R}{n}} + \sqrt{\delta} \\ &\leq (1 + \frac{2L}{\alpha})\sqrt{\delta} + \frac{2\sigma}{\alpha} \sqrt{\frac{2 \ln 2(2 + \eta)R}{n}} \end{aligned} \quad (74)$$

□

B. Implementation details

B.1. Dataset and baseline

Dataset. Following (Zafari et al., 2026), we evaluate denoising performance under AWGN and Poisson noise on Set11 (Zhang et al., 2017), Set13 (Zeyde et al., 2010) (192×192 crops), and Kodak24 (Kodak, 1991). Specifically, the Set11 dataset consists of 11 grayscale images (C.man, House, Peppers, Starfish, Monarch, Airplane, Parrot, Barbara, Boats, Pirate, and Couple), where seven images have a resolution of 256×256 , and the remaining four images have a resolution of 512×512 . Set13 dataset consists of 13 images (Baboon, Barbara, Bridge, Coastguard, Comic, Face, Flowers, Foreman, Man, Monarch, Peppers, PPT3, and Zebra), where each image is center-cropped into 192×192 patches. Kodak dataset consists of 24 images, each with a resolution of 512×768 . For super-resolution and inpainting, we follow (Ulyanov et al., 2020) using Set5 (Bevilacqua et al., 2012) and other images as in their original papers.

Baseline. For denoising, we compare our approach against representative zero-shot solvers, including both traditional and learning-based methods. All baselines are dataset-free and require no external training. For non-learning methods, we include JPEG2K and BM3D. For learning-based methods, we evaluate Deep Image Prior (DIP) (Ulyanov et al., 2020), Deep Decoder (DD) (Heckel & Hand, 2019), Zero-Shot Noise2Self (ZS-N2S) (Batson & Royer, 2019), Self2Self (S2S) (Quan et al., 2020), Zero-Shot Noise2Noise (ZS-N2N) (Mansour & Heckel, 2023), and Zero-Shot Neural Compression Denoiser (ZS-NCD) (Zafari et al., 2026). The performance comparison is summarized in Table 1. The baseline results were obtained using their official implementations. For alternative architectures, we employ their optimal results or use the better result between our reproduction and the officially reported results from their original papers/codes.

For noisy super-resolution, we compare against representative classical and zero-shot baselines, including Bicubic interpolation, No-Prior optimization, BM3D, BM3D+Bicubic, DIP, BM3D+LapSRN, and ZS-NCD+LapSRN (Lai et al., 2017). For inpainting, DIP serves as the primary zero-shot baseline. For text removal, we additionally compare with Shepard-Net (Ren et al., 2015). For region inpainting, we include a representative generative-model-based approach, Local-Global GAN (Iizuka et al., 2017), to assess performance against methods with explicit generative priors.

B.2. Model architecture

We summarize the detailed architecture of the employed implicit codec in Table 7. For the auto-regressive entropy model, we use 32 contextual elements as input and adopt a three-layer network composed of linear (or 1×1 convolutional) layers with GELU activations. The network follows the configuration: $(32 \times 32) \rightarrow \text{GELU} \rightarrow (32 \times 32) \rightarrow \text{GELU} \rightarrow (32 \times 32)$.

For the modulation network f_θ , we set the number of synthesis layers to $L_t = 4$ and use $L = 7$ multi-resolution latent components with an 8×8 upsampling kernel. The network maps the upsampled latent features through a lightweight MLP with dimensions: $(7 \times 48) \rightarrow \text{GELU} \rightarrow (48 \times 3) \rightarrow \text{GELU} \rightarrow (3 \times 3)$.

The randomized network takes pixel coordinates as input and consists of $L_t = 4$ layers. Coordinates are first embedded using Fourier-feature positional encoding, yielding a 24-dimensional representation, which is concatenated with the original coordinates to form a 26-dimensional input. The network architecture is: $(26 \times 32) \rightarrow \text{GELU} \rightarrow ([32 + 3 + 3] \times 24) \rightarrow \text{GELU} \rightarrow ([24 + 3 + 3 + 3] \times 16) \rightarrow ([16 + 3 + 3 + 3 + 48] \times 3) \rightarrow \text{Tanh}$, where layer-wise modulation features produced by the modulation network are concatenated in a rewind fashion. This design enables efficient structure-aware modulation while keeping the network lightweight.

B.3. Fourier initializations for random network

We adopt the Fourier-based initialization (Shi et al., 2024) to facilitate the identification of a *Lottery Prior* that mitigates the low-frequency bias of MLPs and preserves informative sign patterns while maintaining stable variance (Zhou et al., 2019; Glorot & Bengio, 2010; He et al., 2016). Concretely, weight matrix $\mathbf{W}^{(i)} \in \mathbb{R}^{d_i \times d_{i-1}}$ of each i layer is reparameterized as $\mathbf{W}^{(i)} = \mathbf{\Lambda}^{(i)} \mathbf{B}^{(i)}$, where $\mathbf{\Lambda}^{(i)} \in \mathbb{R}^{d_i \times M}$ denotes the coefficient matrix, and $\mathbf{B}^{(i)} \in \mathbb{R}^{M \times d_{i-1}}$ contains M Fourier bases, defined by distinct frequency–phase pairs. We employ P phase shifts and both low- and high-frequency components, yielding $M = P(F_{\text{high}} + F_{\text{low}})$. Throughout this paper, we set $P = 32$, $F_{\text{high}} = 256$, and $F_{\text{low}} = 32$.

B.4. Training details

We select λ empirically according to the degradation level ($\mathbf{A}$ is given), with more severe degradations requiring stronger rate constraints (higher compression). In principle, degradation levels can be estimated using standard blind restoration techniques (You & Dragotti, 2024).

For denoising, we adopt the Lagrangian coefficient search strategy of (Zafari et al., 2026). Under the AWGN model $\mathbf{y} = \mathbf{x} + \mathbf{z}$, we have $\frac{1}{n} \mathbb{E} \|\mathbf{y} - \hat{\mathbf{x}}\|_2^2 = \sigma^2 + \frac{1}{n} \mathbb{E} \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \frac{2}{n} \mathbb{E} [\mathbf{z}^\top (\mathbf{x} - \hat{\mathbf{x}})]$. In effective denoising regimes the cross term is negligible, so the residual energy $\frac{1}{n} \|\mathbf{y} - \hat{\mathbf{x}}\|_2^2$ is close to the noise variance. We therefore choose λ by matching the residual to an estimate of σ^2 .

For noisy super-resolution and inpainting, we find that the method is robust to the choice of λ , and thus a single λ is used for each degradation level.

We report λ in later visualization section C and data-points section E. Other hyperparameters are reported in Table 7.

B.5. Ensemble of priors

A simple way to aggregate priors (obtained from different random seeds and mask ratios) is to directly average them, which achieves state-of-the-art performance for compression-based zero-shot denoising and noisy super-resolution.

We also provide a linear averaging approach for improved denoising performance. Given a set of candidate reconstructions $\{\hat{\mathbf{x}}_i\}_{i=1}^K$, we evaluate their measurement consistency with respect to the noisy observation $\mathbf{y}$. This is quantified by the mean squared error in the denoising case: $e_i = \|\hat{\mathbf{x}}_i - \mathbf{y}\|_2^2$.

We then compute aggregation weights via a gating mechanism,

$$\gamma_i = \frac{\exp(-\tau (e_i - \min_j e_j))}{\sum_{j=1}^K \exp(-\tau (e_j - \min_\ell e_\ell))}, \quad (75)$$

where the temperature parameter is set to $\tau = 50$ to control the sharpness of the selection. This yields a prior-aggregated estimate $\bar{\mathbf{x}} = \sum_{i=1}^K \gamma_i \hat{\mathbf{x}}_i$. Finally, a data-anchoring proximal step balances fidelity to the noisy observation and the aggregated prior:

$$\mathbf{x}^* = \frac{\mathbf{y} + \lambda \bar{\mathbf{x}}}{1 + \lambda}, \quad (76)$$

where $\lambda = 30$ controls the effect of the ensemble prior. For more details, we refer to our project page. Note that we use simple linear aggregation for denoising, which is sufficient to achieve strong performance; more advanced aggregation strategies are left for future work. To show the effect of the ensemble, we provide a visualization in Fig. 7.

B.6. Pseudocode for the algorithm

This section provides a detailed learning and inference algorithm of Lottery Prior, as shown in Algorithm 1 and Algorithm 2.

Table 7. Hyper-parameter settings.

Hyper parameter	Initial values	Final values
Quantization – Stage I		
Number of learning steps	$5 * 10^4$	
Learning rate β	10^{-2}	0
Scheduler for learning rate	Cosine scheduler	
Temperature T for soft rounding	0.3	0.1
Noise strength α for Kumaraswamy noise	2.0	1.0
Scheduler for Soft-rounding and Kumaraswamy noise	Linear scheduler	
Quantization – Stage II		
Number encoding steps	10^4	
Learning rate	10^{-4}	10^{-8}
Decay learning rate if loss has not improved for this many steps	40	
Decay factor	0.8	
Temperature T for soft rounding	10^{-4}	
Architecture – Latent modulations	Values	
Number of latent vectors L	7	
Initialization of $\mathbf{z}$	0	
Architecture – ModNet	Values	
Output channels of the 1×1 convolutions	{48, 3}	
Number of 3×3 residual convolutions	2	
Architecture – Entropy model		
Output channels of 1×1 convolutions	{32, 32, 2}	
Activation function	GELU	
Log-scale of Laplace is shifted before exp	4	
Scale parameter of Laplace is clipped to	$[10^{-2}, 150]$	
Architecture – randomized network	Values	
Output dimensions of MLP layers	{32, 24, 16, 3}	
Mask ratios	{0.2, 0.8} with 0.05 intervals	
Initialization of MLP	FFN initialization	
FFN initialization phase number P	32	
FFN initialization low/high frequency number F	32/256	
Initialization of score matrix $\mathbf{P}$	Kaiming uniform initialization	
Learning rate α for $\mathbf{P}$	0.1	
Scheduler	Cosine scheduler	

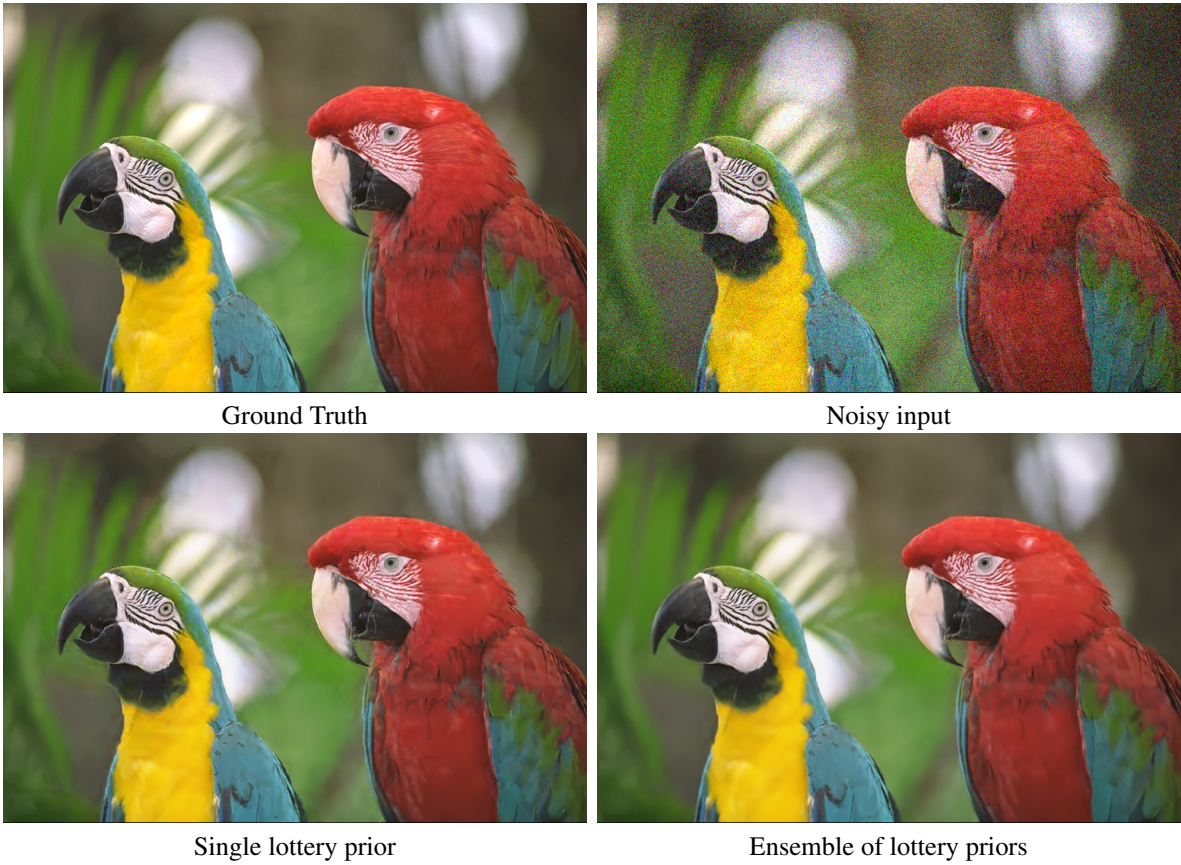


Figure 7. Denoising on Kodak under AWGN ($\sigma = 25$). Prior ensembling improves restoration and effectively suppresses artifacts.

Algorithm 1 Zero-shot learning stage of the Lottery Prior

Input: Observed measurement $\mathbf{y}$, degradation operator $\mathbf{A}$, random seed e ,

learning rates α for mask scores ($\mathbf{P}$),

learning rates β for modulations ($\mathbf{z}$), and networks (θ and ψ),

cosine scheduler for learning rate for Stage I and II: $\mathcal{C}_1, \mathcal{C}_2$,

linear scheduler for soft-rounding temperature and Kumaraswamy noise strength for Stage I and II: l_1, l_2 ,

Output: $\hat{\mathbf{Z}}, \theta, \tau$.

```

1:  $\mathbf{w}_0 = \mathbf{A}\mathbf{B}, \mathbf{P} \sim \mathcal{U}_k$ , // Initialization for  $g(\cdot)$  and  $\mathbf{P}$  based on  $e$ 
2: for the  $i$ -th step within the Stage I do
3:    $\tau = h(\mathbf{P}), g = g_{\tau \odot \mathbf{w}_0}$  // Configure network by masking  $k\%$  weights based on  $\mathbf{P}$ 
4:    $\hat{\mathbf{Z}} = \mathcal{S}_T(\mathbf{Z}) + \mathbf{u}_{kum}^{kum}$ , // Quantization-aware training
5:    $\hat{\mathbf{x}} = g_{\tau \odot \mathbf{w}_0}(f_\theta(\hat{\mathbf{Z}}))$ , // Modulate the reconstruction
6:    $\mathcal{L} = D(\mathbf{y}, \mathbf{A}\hat{\mathbf{x}}) + \lambda R_\psi(\hat{\mathbf{Z}})$ , // Compute the RD cost loss function
7:    $\mathbf{P} \leftarrow \mathbf{P} - \alpha \nabla_{\mathbf{P}} \mathcal{L}$ , // Update the scores for mask
8:    $\mathbf{Z} \leftarrow \mathbf{Z} - \beta \nabla_{\mathbf{Z}} \mathcal{L}$ , // Update latent modulations
9:    $\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}$ , // Update modulation net
10:   $\psi \leftarrow \psi - \beta \nabla_{\psi} \mathcal{L}$ , // Update ARM
11:   $\alpha = \mathcal{C}_1(\alpha, i), \beta = \mathcal{C}_1(\beta, i)$  // Update learning rate
12:   $T = l_1(T, i), u_{kum} = l_1(u_{kum}, i)$  // Update noise strength
13: end for
14: for the  $i$ -th step within the Stage II do
15:   $\tau = h(\mathbf{P}), g = g_{\tau \odot \mathbf{w}_0}$  // Mask  $k\%$  weights based on updated  $\mathbf{P}$ 
16:   $\hat{\mathbf{Z}} = Q(\mathbf{Z})$ , // Hard rounding
17:   $\hat{\mathbf{x}} = g_{\tau \odot \mathbf{w}_0}(f_\theta(\hat{\mathbf{Z}}))$ , // Modulate the reconstruction
18:   $\mathcal{L} = D(\mathbf{y}, \mathbf{A}\hat{\mathbf{x}}) + \lambda R_\psi(\hat{\mathbf{Z}})$ , // Compute the RD cost loss
19:   $\mathbf{P} \leftarrow \mathbf{P} - \alpha \nabla_{\mathbf{P}} \mathcal{L}$ , // Update the scores for mask
20:   $\mathbf{Z} \leftarrow \mathbf{Z} - \beta \nabla_{\mathbf{Z}} \mathcal{L}$ , // Update latent modulations
21:   $\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}$ , // Update modulation net
22:   $\psi \leftarrow \psi - \beta \nabla_{\psi} \mathcal{L}$ , // Update ARM
23:   $\alpha = \mathcal{C}_2(\alpha, i), \beta = \mathcal{C}_2(\beta, i)$  // Update learning rate
24:   $T = l_2(T, i), u_{kum} = l_2(u_{kum}, i)$  // Update noise strength
25: end for
26:  $\Omega = \{e, \hat{\mathbf{Z}}, \theta, \tau\}$ , // Save the parameter set
    
```

Algorithm 2 Restoration stage of *Lottery Prior*

Input: Random seed e , and a well-trained set $\{\hat{\mathbf{Z}}, \theta, \tau\}$.

Output: Reconstruction of image $\hat{\mathbf{x}}$.

```

1:  $\mathbf{w}_0 = \mathbf{A}\mathbf{B}, g = g_{\tau \odot \mathbf{w}_0}$  // Configure randomized network with a given seed  $e$ 
2:  $\hat{\mathbf{x}} = g_{\tau \odot \mathbf{w}_0}(f_\theta(\hat{\mathbf{Z}}))$ , // Reconstruct the source
    
```

C. Visualization

C.1. Super-resolution with noisy observation

We visualize $\times 4$ ($\lambda = 7e - 2$) and $\times 8$ ($\lambda = 4e - 1$) super-resolution results on Set5 under additive white Gaussian noise ($\sigma = 15$) in Fig. 8 and Fig 9, respectively. Our method achieves high reconstruction quality and produces clean results comparable to the BM3D+LapSRN pipeline, which uses a supervised super-resolution network.

C.2. Inpainting

For text-removal inpainting, we employ $\lambda = 8e - 4$. For random-mask inpainting, we employ $\lambda = 1e - 4$. We visualize our inpainting results with different mask ratios (ρ) on Set5 in Fig. 10. Our approach yields cleaner textures and fewer artifacts, especially under the heavier masking setting. We also evaluate our approach on region inpainting in Figs 11 ($\lambda = 5e - 3$) and 12 (λ reported in the figure), highlighting the effectiveness of our approach.

C.3. Denoising

We visualize denoising results on Kodak under additive white Gaussian noise ($\sigma = 15, 25$) in Figs. 13 and 14, respectively. We further present Kodak results under Poisson noise ($\alpha = 15$) in Fig. 15, and Set11 results under additive white Gaussian noise ($\sigma = 25$) in Fig. 16. Our approach achieves high fidelity and produces clean results comparable to the supervised Restormer approach.

D. Ablation studies and more experiments

D.1. Ablation study

Model component. We conduct ablation studies over each component (Kodak01 with AWGN $\sigma = 25$). We sequentially remove each component from the proposed Lottery Prior, with results reported in Table 8. The ablation over the ensemble mechanism demonstrates the importance of flexible priors generated from our Lottery Priors. Ablation on the randomized network without the modulation network shows the pure effect of the entropy regularizer and the network priors, revealing that the proposed modulation mechanism can further enhance performance. We also found that lightweight ARM here can result in slightly worse performance.

Table 8. Ablation study on different mechanisms, conducted on the Kodak01 image using an ARM-32 model and a mask ratio of 50%.

Model Variant	Denoising PSNR
<i>Lottery Prior</i> scheme	29.15 dB
⇒ Single prior w/o ensemble (ratio=0.5)	27.18 dB
⇒ randomized network w/o modulation network	26.78 dB
⇒ cheaper ARM with dimension 24	26.89 dB
⇒ Single prior w/o entropy model	Not work

Furthermore, we conducted ablations on optimizer choice (Adam/AdamW/RMSProp) and architectural configurations (backbone width) to assess sensitivity, using the first three images of Set11 for denoising ($\sigma = 15$, mask ratio 0.5).

Optimizers. As shown in Table 9, across the optimizers (Adam, AdamW, RMSProp), performance varies by only 0.12 dB, with only minor differences in convergence speed, confirming that the method is robust to optimizer choice. We therefore adopt the default Adam optimizer.

Architecture. We vary the random network backbone width ($0.5 \times / 1 \times / 2 \times$). As shown in Table 10, narrower models slightly underfit, while wider models yield only marginal gains or may overfit (across a $4 \times$ range in backbone width, performance varies by only 0.06 dB, indicating that the rate constraint, not the specific architecture, is the primary regularizer, consistent with the minimum description length principle and our theoretical framework. From an optimization perspective,

an appropriate model size is preferred, neither too small nor excessively large. We want to highlight that under the randomized network setting, different architectures can induce a family of random codec priors; judiciously combining multiple such priors may further improve performance, which we leave for future work.

Table 9. Ablation over different optimizers.

Lottery Prior	Adam	AdamW	RMSprop
PSNR(dB)	32.60	32.59	32.48
SSIM	0.8834	0.8828	0.8792

Table 10. Ablation over the width of random network.

Width of each layer	$\times 0.5$	$\times 1$	$\times 2$	Combine Priors
PSNR(dB)	32.54	32.60	32.59	32.90
SSIM	0.8826	0.8834	0.8820	0.8872

D.2. Additional experiments over super-resolution tasks

Additional metrics for the super-resolution task are reported in Table 11.

Table 11. Average PSNR / MS-SSIM / LPIPS on Set5 with AWGN ($\sigma = 15$) for the super-resolution tasks.

	4 $\times$ super-resolution	8 $\times$ super-resolution
Methods	PSNR / MS-SSIM / LPIPS	PSNR / MS-SSIM / LPIPS
No prior	20.10/0.7125/0.5682	18.66/0.6102/0.6037
Bicubic	23.32/0.8129/0.6129	20.91/0.7158/0.7433
BM3D+Bicubic	24.99/0.9078/0.4316	21.62/0.7795/0.6296
Deep image prior	21.37/0.7610/0.5064	19.55/0.6460/0.6443
Lottery prior	25.87/0.9182/0.3042	22.07/0.7951/0.5180
LapSRN	24.21/0.8465/0.4158	21.30/0.7418/0.5787
BM3D+LapSRN	25.17 /0.9096/0.3392	21.66/0.7803/0.5189

D.3. Additional experiments over modality and foundation models

Demosaicing and microscopy image. We include additional comparisons with a representative foundation model, RAM. We evaluate on Mouse Nuclei fluorescence microscopy denoising (Buchholz et al., 2020)(Table 12) and natural-image demosaicing (Table 4), both unseen during RAM’s pretraining. For fine-tuning, RAM uses an additional N=10 measurements, following Table 3 of (Terris et al., 2025a). Our experiments show that RAM achieves strong performance without fine-tuning across unseen tasks and modalities. In particular, it is competitive with BM3D on unseen-modality denoising and also achieves strong results on the unseen task of demosaicing, benefiting from its incorporation of forward-operator knowledge and extensive pretraining across diverse modalities and restoration tasks. In contrast, Lottery Prior does not rely on any external data or pretraining, yet still achieves strong performance on both tasks, outperforming pretrained RAM by an average of 0.85 dB on microscopy denoising and 0.84 dB on demosaicing, which validates the generality of the proposed method. At the same time, RAM improves substantially after few-shot self-supervised fine-tuning (N=10), highlighting the advantage of pretrained priors combined with fine-tuning. We note that RAM is not directly comparable to our setting, as it relies on extensive supervised pretraining across diverse datasets and relevant tasks, whereas our method is purely dataset-free and instance-level. We therefore present these results to calibrate our method against pretrained and finetuned foundation-model priors, highlighting the different assumptions in prior knowledge and training resources.

MRI. To assess our method in a medical imaging setting, we conduct experiments on fastMRI (Zbontar et al., 2018; Knoll et al., 2020) single-coil knee reconstruction (320x320-cropped Knee-file1000000 as an example) under 4 $\times$ and 8 $\times$ acceleration. The undersampled measurements follow $y = M\mathcal{F}(x) + n$, where $\mathcal{F}$ denotes the Fourier transform, M the sampling mask, n the acquisition noise ($n = 0$ in our experiments). In this setting, the complex-valued image is

Table 12. Modality adaptation experiments of the foundation models and Lottery Prior on Mouse Nuclei fluorescence microscopy image denoising (first 5 samples from the dataset for validation, and randomly sampled 10 from the remaining as a fine-tuning set). Reported values are the average PSNR, and Lottery Prior is averaged with 3 priors, each with 10k steps.

Noise level	BM3D	Lottery Prior	RAM (Pretrained)	RAM (Finetuned)
$\sigma = 15$	33.91	34.13	33.89	34.98
$\sigma = 25$	29.85	30.42	29.57	29.91

parameterized by a single lottery prior and optimized under a k -space data-consistency objective with a rate constraint. The PSNR results are reported in Table 13, showing that the proposed Lottery Prior can extend to medical imaging inverse problems beyond natural image settings. Our method achieves consistent improvement over DIP (+0.3 dB at $4\times$, +1.26 dB at $8\times$, demonstrating that the compression-based prior generalizes to MRI reconstruction. We note that both dataset-free approaches (DIP and ours) naturally have a performance ceiling compared to methods that exploit domain-specific training data or imaging physics priors. Bridging this gap through hybrid approaches (e.g., integrating Lottery Prior as a regularizer within physics-informed unrolled networks (Huang et al., 2023)) is a compelling direction for future work.

Table 13. Experiment over MRI reconstruction (Knee) from $4\times$ and $8\times$ undersampled k -space measurement.

PSNR performance	$4\times$	$8\times$
Deep image prior	26.15	23.92
Lottery prior	26.45	25.18

D.4. Complexity analysis

We report the detailed runtime and computational complexity in Table 6 (first 3 images of Set11 for denoising with $\sigma = 15$, mask ratio 0.5 under the same configuration of Table 1).

Latency and complexity. To ensure fairness across different optimization schedules, we measure GPU runtime and complexity under a fixed budget of 1000 iterations. Compared to ZS-NCD, which trains a 0.4M-parameter network over overlapping patches (officially reported 40 minutes per image), our method is significantly more efficient due to its lightweight design and avoidance of extensive patch-wise redundancy. Compared to DIP, which relies on a heavy multi-scale architecture (2.2M parameters), we consider its lightweight variant for fairness (1–2 minutes). In contrast, our model is highly compact (0.7–3.3k parameters), leading to substantially lower per-iteration cost and better performance.

Learning steps. We further analyze performance across optimization steps (reported in Table 14) and observe that satisfactory results are achieved within 5–10k iterations, while longer runs provide additional but diminishing gains, indicating a favorable time–performance trade-off that can be flexibly adjusted based on the available computational budget.

Table 14. Denoising performance across various zero-shot learning steps.

Learning steps	500	1000	3000	5000	7000	10000	20000	30000	50000
PSNR	31.32	31.75	32.19	32.34	32.39	32.42	32.49	32.55	32.60
SSIM	0.8649	0.8710	0.8766	0.8793	0.8798	0.8802	0.8817	0.8825	0.8834

Prior aggregation. Our framework also supports flexible prior aggregation: combining a few randomized priors (e.g., 5 mask ratios, ~ 10 – 20 minutes) improves denoising, while a single prior (5–10k steps, ~ 2 minutes) suffices for structured tasks such as inpainting, avoiding over-smoothing, and achieving state-of-the-art performance. We note that the ensemble-based refinement introduces additional computational overhead. However, this component is optional and not required by the core method. In practice, a single run already achieves competitive (and for some tasks, such as inpainting, state-of-the-art) performance.

To summarize: in terms of total wall-clock time for a single image: ZS-NCD requires ~ 40 min (20K steps), DIP requires $\sim 1 - 2$ min (2-4K steps), and a single Lottery Prior requires $\sim 1 - 2$ min (5-10K steps at much lower per-step cost). For

ensemble refinement with 5 priors, total time increases to $\sim 5 - 10$ min, which remains competitive with ZS-NCD while achieving superior performance.

E. Detailed datapoints

We report detailed results for each experiment. For baselines, we use the better result between our reproduction and the officially reported results from their original papers/codes, ensuring a fair comparison to prior methods.

Denoising-Set11 For denoising on Set11, results under AWGN are reported in Table 15 ($\lambda = 0.004/0.011$ for $\sigma = 15/25$), while results under Poisson noise are shown in Table 16 ($\lambda = 0.04/0.025$ for $\alpha = 15/25$).

Denoising-Set13 For denoising on Set13, results under AWGN are reported in Table 17 ($\lambda = 0.0015/0.004$ for $\sigma = 15/25$), while results under Poisson noise are shown in Table 18 ($\lambda = 0.013/0.008$ for $\alpha = 15/25$).

Denoising-Kodak For denoising on Kodak, results under AWGN ($\lambda = 0.0015/0.0035$ for $\sigma = 15/25$) are reported in Table 19, while results under Poisson noise are shown in Table 20 ($\lambda = 0.011/0.0075$ for $\alpha = 15/25$).

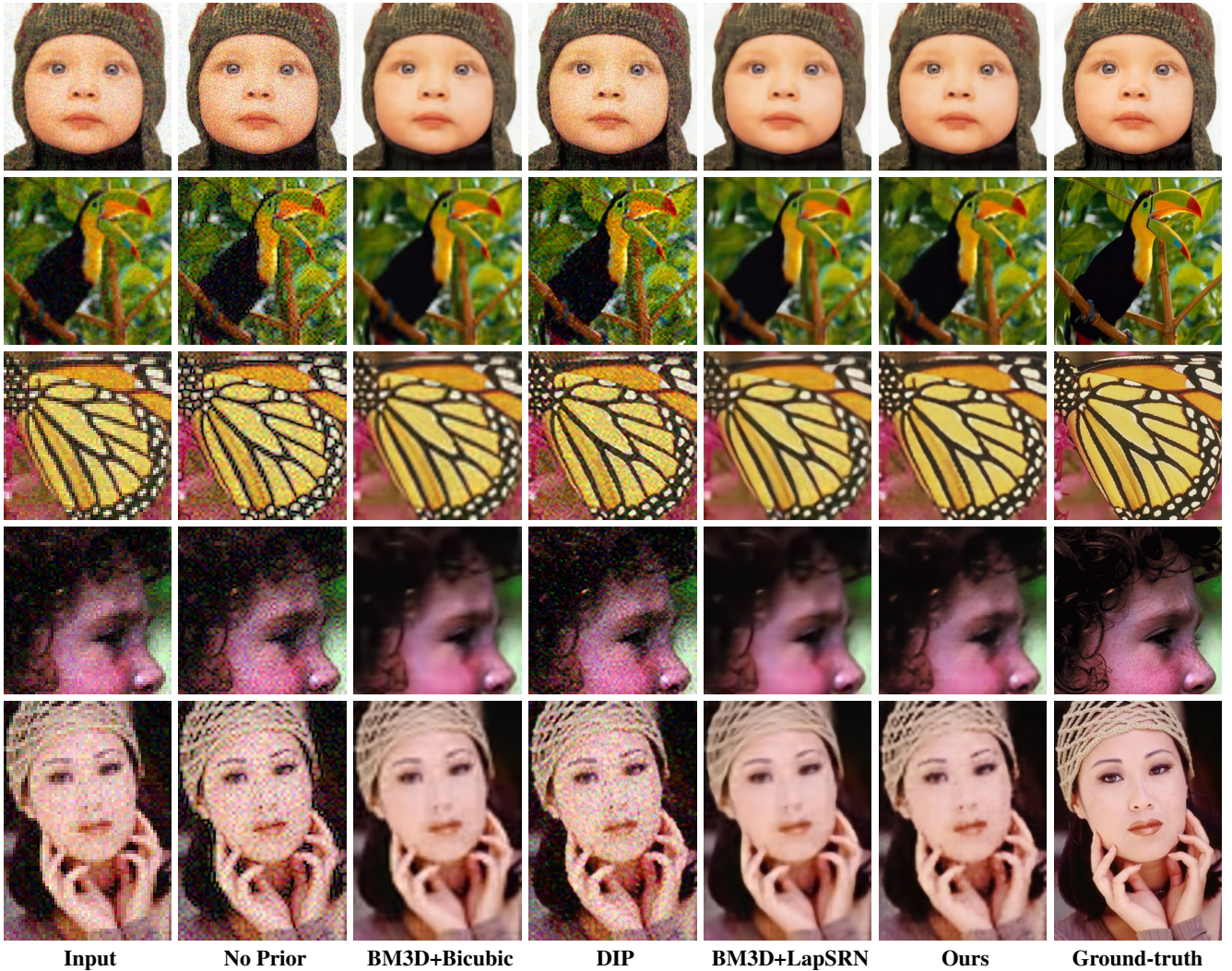


Figure 8. $\times 4$ super-resolution results on Set5 under additive white Gaussian noise ($\sigma = 15$). Our method achieves high reconstruction quality and produces clean results comparable to the BM3D+LapSRN pipeline, which uses a supervised super-resolution network.

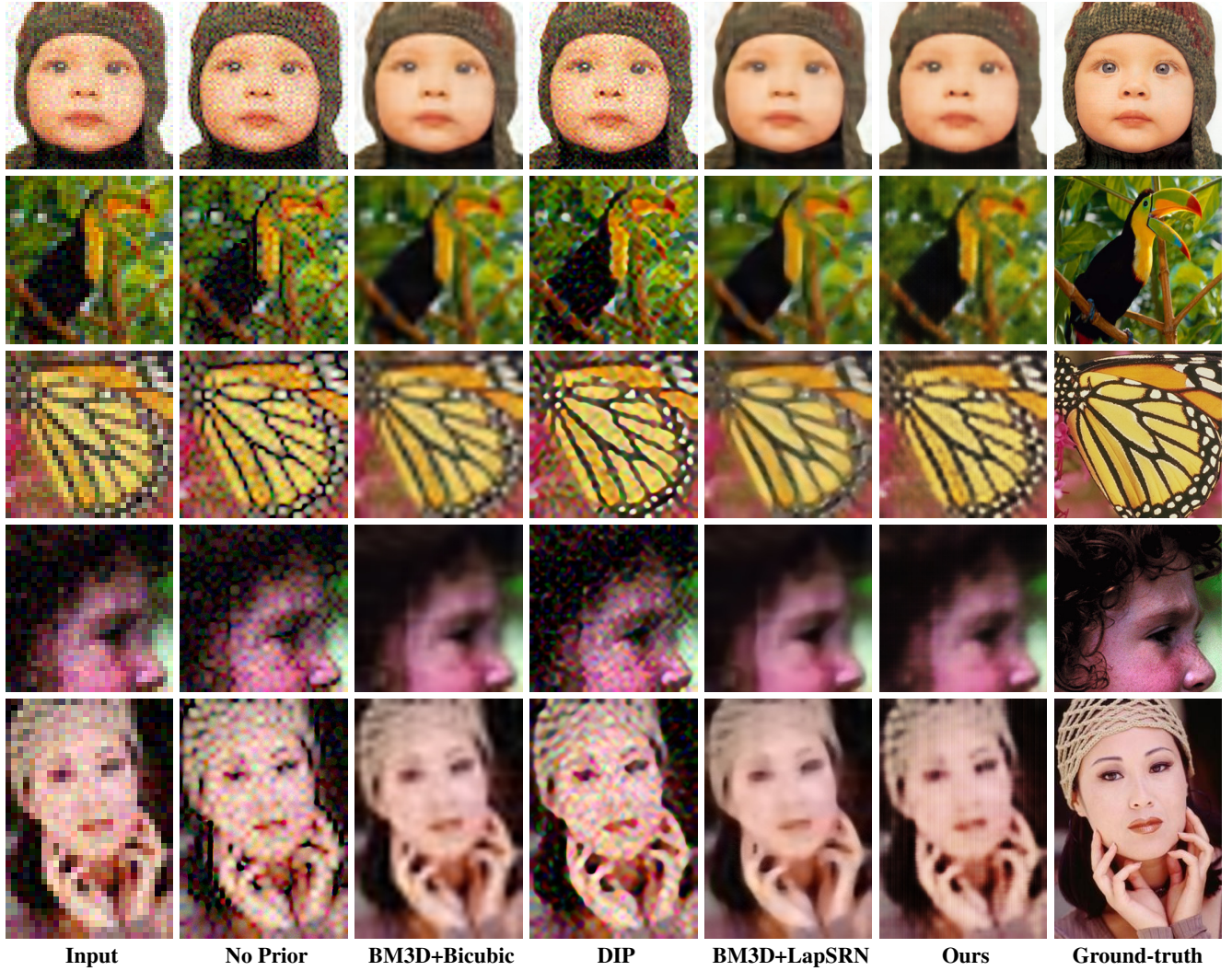


Figure 9. $\times 8$ super-resolution results on Set5 under additive white Gaussian noise ($\sigma = 15$). Our method achieves high reconstruction quality and produces clean results comparable to the BM3D+LapSRN pipeline, which uses a supervised super-resolution network.

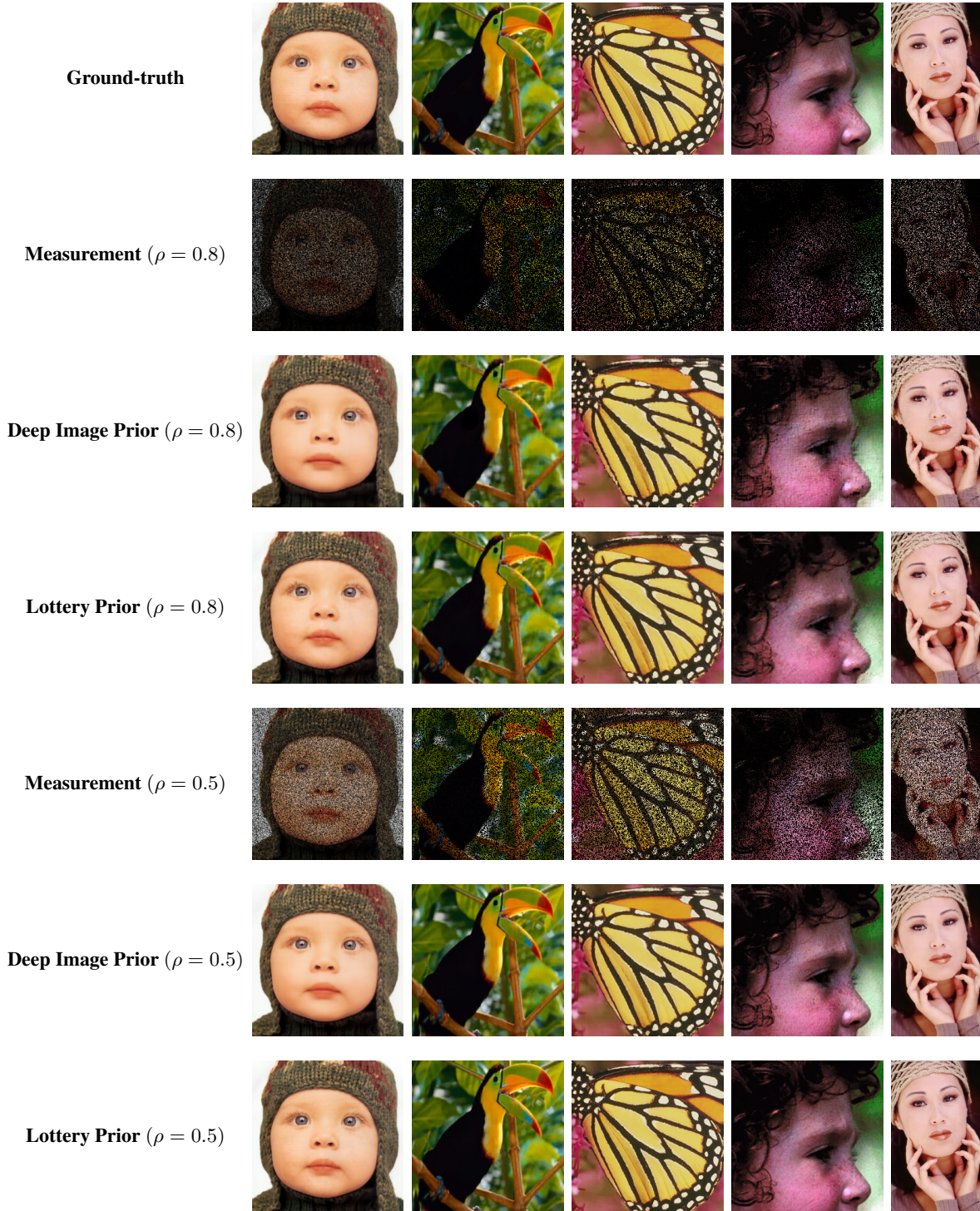


Figure 10. Inpainting results with different mask ratios (ρ) on Set5. Our *Lottery Prior* yields cleaner textures and fewer artifacts, especially under the heavier masking setting.

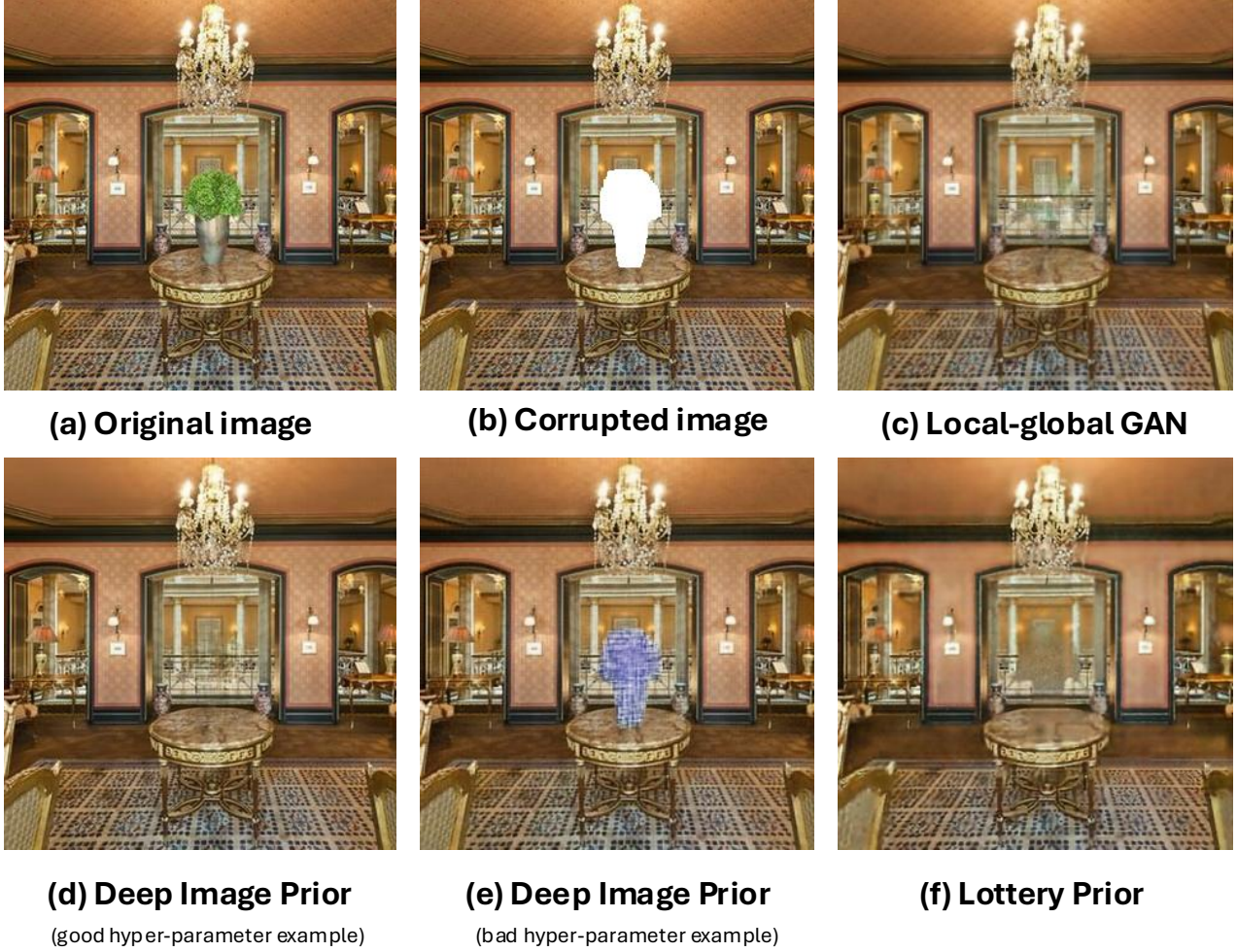


Figure 11. Region inpainting: vase. Deep Image Prior (DIP) can inpaint large missing regions despite the absence of external training, achieving results comparable to learned generative methods such as (Iizuka et al., 2017). However, its performance is sensitive to architectural choices and hyperparameters, as illustrated by the variation across (d) and (e). In contrast, *Lottery Prior*, while not designed for region-level semantic generation as in GAN-based methods, is still able to complete missing regions by exploiting contextual information from entropy modeling and the global fitting ability of an overfitted codec. Notably, this behavior is achieved with a substantially lighter architecture, suggesting that randomized masked networks can serve a role analogous to DIP as an implicit structural prior, with further gains possible by increasing model capacity.

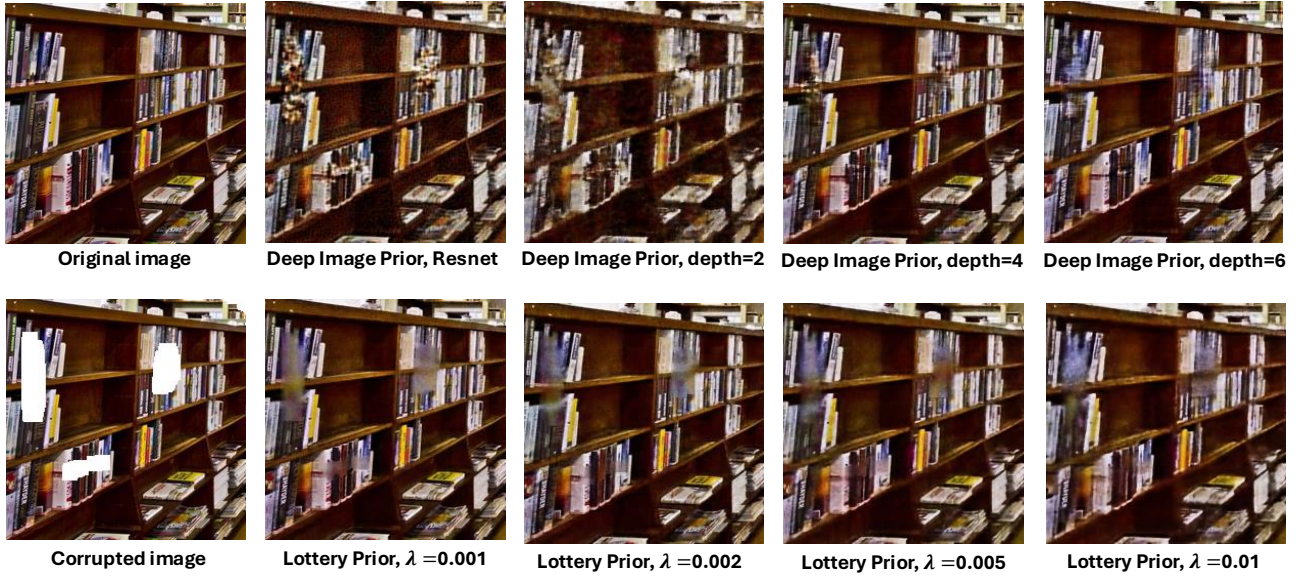


Figure 12. Region inpainting: library. The library scene poses a challenging inpainting task due to its complex structure. Deep Image Prior here requires very deep architectures and carefully tuned hyperparameters to achieve satisfactory results; otherwise, reconstructions degrade noticeably. In contrast, *Lottery Prior* controls reconstruction behavior via the rate constraint: larger λ enforces stronger compression, improving consistency in the surrounding regions but suppressing background details. Interestingly, this behavior is similar to early stopping in DIP, complex scenes require earlier stopping, which leads to underfitting of background structures. From a compression perspective, this corresponds to operating at a higher compression rate, resulting in background smoothing. Motivated by DIP, increasing network capacity is expected to further improve inpainting quality for such complex scenes.

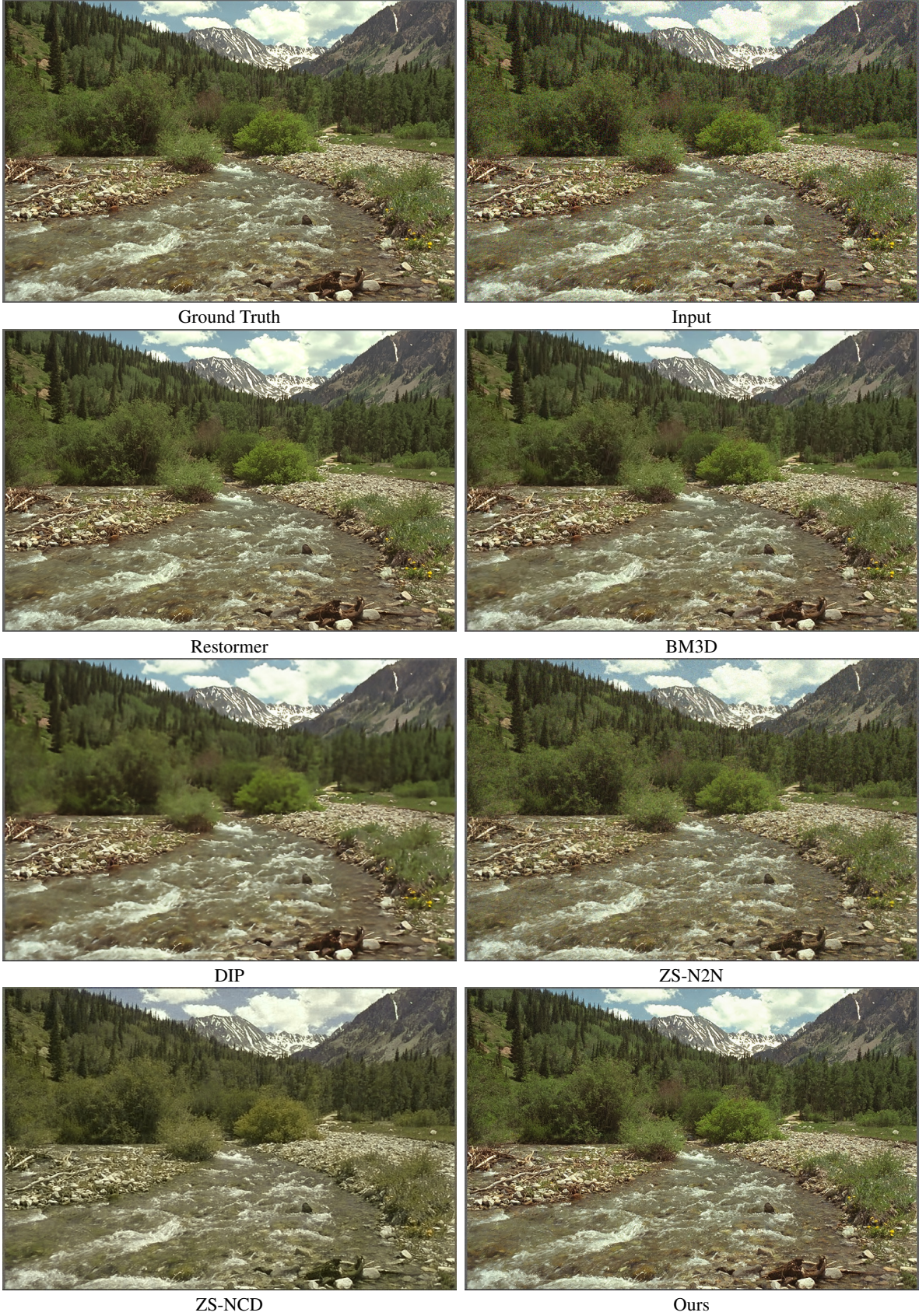


Figure 13. Denoising on Kodak under additive white Gaussian noise ($\sigma = 15$). Our approach achieves high fidelity and produces clean results comparable to the supervised Restormer approach.



Figure 14. Denoising on Kodak under additive white Gaussian noise ($\sigma = 25$). Our approach achieves high fidelity and produces clean results comparable to the supervised Restormer approach.

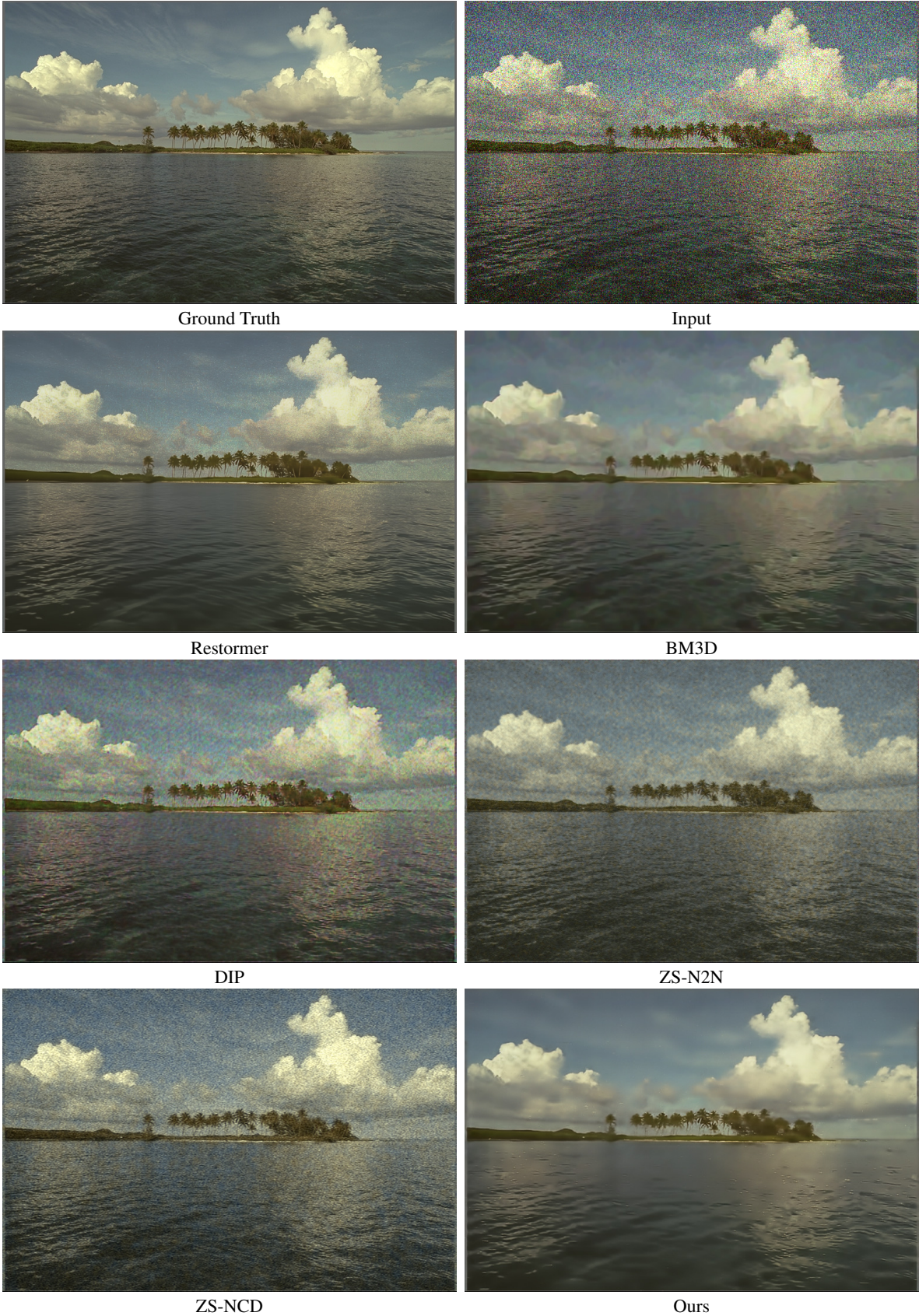


Figure 15. Denoising on Kodak under Poisson noise ($\alpha = 15$). Our approach achieves high fidelity and produces clean results comparable to the supervised Restormer approach.



Figure 16. Denoising on Set11 under additive white Gaussian noise ($\sigma = 25$). Our approach achieves high fidelity and produces clean results comparable to the supervised Restormer approach.

Lottery Prior: Randomized Neural Compression for Zero-Shot Inverse Problems

Table 15. Set11 Denoising performance comparison under AWGN (PSNR and SSIM).

σ	Method	256 $\times$ 256							512 $\times$ 512				Average
		C.man	House	Peppers	Starfish	Monarch	Airplane	Parrot	Barbara	Boats	Pirate	Couple	
15	BM3D	31.84/0.8974	34.94/0.8870	32.82/0.9118	31.16/0.9082	31.92/0.9409	31.09/0.9034	31.47/0.9032	33.04/0.9253	32.12/0.8604	31.94/0.8726	32.11/0.8795	32.22/0.8991
	JPEG2K	27.12/0.7474	29.48/0.7621	27.96/0.7907	26.75/0.8077	26.74/0.8166	26.58/0.7664	27.30/0.7778	26.76/0.7690	27.87/0.7390	27.92/0.7471	27.44/0.7449	27.45/0.7699
	DIP	27.94/0.7417	31.39/0.8111	29.80/0.8273	29.58/0.8605	29.93/0.8767	28.14/0.8047	28.37/0.7794	27.65/0.7538	29.48/0.7798	29.27/0.7817	28.65/0.7727	29.11/0.7990
	DD	29.41/0.8099	32.83/0.8406	26.97/0.8488	29.39/0.8739	30.01/0.8957	26.44/0.8228	29.32/0.8447	24.48/0.7089	29.45/0.7883	29.78/0.8085	29.06/0.7938	28.83/0.8215
	ZS-N2N	30.14/0.8133	32.19/0.8138	30.58/0.8264	29.52/0.8639	30.15/0.8551	29.98/0.8298	30.19/0.8290	27.70/0.7772	30.06/0.7900	30.06/0.7957	29.59/0.7913	30.01/0.8169
	ZS-N2S	27.66/0.8272	31.08/0.8442	29.46/0.8675	28.83/0.8810	28.77/0.8961	27.34/0.8591	27.67/0.8528	28.75/0.8534	29.52/0.8139	29.41/0.8181	29.60/0.8311	28.92/0.8495
	S2S	20.29/0.6769	32.96/0.8633	23.96/0.8387	25.50/0.8250	30.05/0.9269	28.10/0.8611	20.20/0.7132	30.35/0.8865	27.74/0.7871	29.97/0.8192	25.82/0.7754	26.81/0.8158
	ZS-NCD	30.83/0.8554	34.45/0.8835	32.20/0.8844	31.34/0.8749	31.83/0.8966	30.07/0.8552	30.40/0.8464	31.14/0.8826	31.09/0.8014	30.82/0.8302	30.67/0.8279	31.35/0.8580
	Lottery Prior	31.82/0.8949	34.55/0.8779	32.90/0.9073	31.92/0.9074	32.73/0.9448	31.35/0.9003	31.48/0.8931	32.94/0.9183	32.11/0.8519	31.77/0.8582	31.71/0.8603	32.30/0.8922
	BM3D	29.54/0.8499	32.79/0.8561	30.13/0.8705	28.58/0.8578	29.36/0.9046	28.50/0.8584	28.94/0.8561	30.63/0.8887	29.87/0.8039	29.64/0.8082	29.74/0.8206	29.79/0.8523
	JPEG2K	24.49/0.6976	27.26/0.7269	24.93/0.7206	24.18/0.7167	24.06/0.7561	23.91/0.7126	24.38/0.7162	24.09/0.6825	25.61/0.6577	25.76/0.6566	25.28/0.6534	24.91/0.6997
25	DIP	25.23/0.6043	28.93/0.7545	27.39/0.7579	26.39/0.7777	27.47/0.8169	25.57/0.6983	26.29/0.7409	24.75/0.6356	27.05/0.6843	27.06/0.6857	26.52/0.6847	26.60/0.7128
	DD	27.24/0.7521	30.48/0.8023	25.39/0.7591	26.86/0.8051	27.69/0.8526	24.93/0.7120	27.29/0.7863	23.81/0.6455	27.49/0.7163	27.49/0.7380	27.13/0.7131	26.93/0.7530
	ZS-N2N	27.32/0.7089	29.36/0.7276	27.46/0.7240	26.61/0.7821	27.20/0.7634	27.02/0.7463	27.16/0.7149	25.49/0.6854	27.26/0.6779	27.48/0.6931	26.63/0.6673	27.18/0.7173
	ZS-N2S	26.24/0.7843	29.23/0.8073	27.77/0.8233	27.61/0.8463	27.35/0.8569	25.86/0.8023	26.27/0.7997	26.43/0.7759	28.23/0.7580	27.40/0.7526	27.74/0.7617	27.30/0.7971
	S2S	16.93/0.5998	29.12/0.8275	21.88/0.7666	21.14/0.6974	25.93/0.8606	24.12/0.7350	17.09/0.6069	25.79/0.7980	23.94/0.7061	27.32/0.7403	23.29/0.6979	23.32/0.7306
	ZS-NCD	28.78/0.8237	32.14/0.8547	29.62/0.8406	28.48/0.8134	29.02/0.8494	27.77/0.8126	28.14/0.8007	28.39/0.8192	28.85/0.7444	28.64/0.7630	28.37/0.7648	28.93/0.8079
	Lottery Prior	29.11/0.8360	32.53/0.8513	30.34/0.8681	29.24/0.8626	29.79/0.9054	28.64/0.8556	28.72/0.8337	30.20/0.8695	29.73/0.7881	29.50/0.7943	29.27/0.7976	29.73/0.8420

Table 16. Set11 denoising performance comparison under Poisson noise (PSNR and SSIM).

α	Method	256 $\times$ 256							512 $\times$ 512				Average
		C.man	House	Peppers	Starfish	Monarch	Airplane	Parrot	Barbara	Boats	Pirate	Couple	
15	BM3D	26.64/0.7651	29.39/0.7668	27.13/0.7914	24.93/0.7519	26.32/0.8265	24.79/0.6730	26.26/0.7866	27.24/0.7860	26.82/0.6977	27.07/0.7048	26.67/0.7056	26.66/0.7505
	JPEG2K	21.98/0.6032	24.35/0.6106	22.12/0.6213	21.52/0.5887	21.24/0.6493	20.87/0.5818	22.02/0.6378	21.94/0.5688	23.09/0.5346	23.75/0.5510	23.01/0.5237	22.35/0.5882
	DIP	22.85/0.5382	26.32/0.6528	24.23/0.6138	23.23/0.6696	23.54/0.6875	22.07/0.4938	22.81/0.5723	22.59/0.5503	24.18/0.5533	24.95/0.5811	23.83/0.5362	23.69/0.5863
	DD	24.45/0.6261	27.59/0.7453	23.20/0.7269	23.86/0.7164	24.66/0.7286	22.22/0.6055	24.38/0.6830	22.89/0.6081	24.64/0.6023	25.61/0.6516	24.58/0.5986	24.37/0.6629
	ZS-N2N	24.19/0.5818	25.41/0.5346	24.65/0.6016	23.12/0.6520	23.92/0.6441	23.12/0.5565	23.83/0.5821	23.05/0.5503	24.40/0.5403	24.87/0.5684	23.94/0.5305	24.04/0.5766
	ZS-N2S	24.94/0.7241	27.29/0.7317	25.71/0.7431	24.41/0.7417	25.37/0.7968	23.05/0.7051	24.98/0.7315	22.87/0.6087	26.08/0.6696	25.94/0.6655	25.09/0.6389	25.06/0.7051
	S2S	23.53/0.7325	22.01/0.7409	22.73/0.7300	18.20/0.6010	21.81/0.7813	16.18/0.5010	22.70/0.7304	22.10/0.7261	23.49/0.6529	24.72/0.6782	24.17/0.6843	21.75/0.6872
	ZS-NCD	25.73/0.7660	28.87/0.8015	26.54/0.7745	24.65/0.6988	25.86/0.7791	24.21/0.6568	25.52/0.7356	24.11/0.6562	25.48/0.6510	25.93/0.6705	25.29/0.6552	25.65/0.7132
	Lottery Prior	25.54/0.6940	29.19/0.7886	26.92/0.7759	25.02/0.7490	26.11/0.8070	25.05/0.7437	25.96/0.7677	25.81/0.7396	25.89/0.6419	26.56/0.6820	25.76/0.6702	26.16/0.7327
	BM3D	22.69/0.5154	22.82/0.4765	22.74/0.5930	22.11/0.6947	23.44/0.7213	19.60/0.3788	23.05/0.5991	23.09/0.6508	22.77/0.5123	23.89/0.5979	23.45/0.5753	22.70/0.5741
25	JPEG2K	22.54/0.6267	24.97/0.6773	22.87/0.6076	22.26/0.6378	22.55/0.6641	21.59/0.5649	22.62/0.6373	22.55/0.5976	23.71/0.5685	24.20/0.5801	23.49/0.5566	23.03/0.6108
	DIP	24.21/0.5976	27.06/0.6553	25.76/0.6945	24.41/0.7312	25.21/0.7384	23.58/0.6290	24.69/0.6608	23.11/0.5903	25.24/0.6130	26.10/0.6528	24.91/0.5998	24.94/0.6512
	DD	25.59/0.6695	28.47/0.7606	24.20/0.7348	25.14/0.7667	26.16/0.8022	23.24/0.6306	25.89/0.7289	23.27/0.6257	25.84/0.6517	26.76/0.6961	25.77/0.6573	25.48/0.7022
	ZS-N2N	25.54/0.6334	27.14/0.6234	25.82/0.6522	24.33/0.7158	25.51/0.7109	24.53/0.6274	25.55/0.6617	24.09/0.6173	25.60/0.6018	26.11/0.6354	25.16/0.5958	25.40/0.6432
	ZS-N2S	26.22/0.7776	27.81/0.7643	26.55/0.7768	24.82/0.7795	26.48/0.8254	24.77/0.7463	25.44/0.7839	23.24/0.6387	27.25/0.7143	27.13/0.7200	26.44/0.6986	26.01/0.7478
	S2S	25.09/0.7572	24.10/0.7398	24.91/0.7733	19.11/0.6491	23.64/0.8226	17.93/0.6279	21.13/0.7692	24.01/0.7860	25.30/0.7058	26.45/0.7232	25.77/0.7360	23.40/0.7355
	ZS-NCD	27.17/0.7635	30.09/0.8109	27.92/0.7932	26.27/0.7600	27.28/0.8093	24.93/0.6116	26.74/0.7551	26.24/0.7393	27.30/0.6993	27.32/0.7192	26.83/0.7123	27.10/0.7431
	Lottery Prior	27.77/0.8004	30.74/0.8219	28.29/0.8185	26.84/0.8042	27.81/0.8596	25.07/0.6364	27.05/0.7950	27.58/0.7914	27.78/0.7236	27.68/0.7259	27.45/0.7309	27.64/0.7734

Lottery Prior: Randomized Neural Compression for Zero-Shot Inverse Problems

Table 17. Set13 denoising performance comparison under AWGN (PSNR and SSIM).

σ	Method	Baboon	Barbara	Bridge	Coastguard	Comic	Face	Flowers	Foreman	Man	Monarch	Peppers	PPT3	Zebra	Average
15	BM3D	28.56/0.7797	33.07/0.9151	30.39/0.8723	30.18/0.8799	28.74/0.9289	30.28/0.7665	29.62/0.9040	35.83/0.9369	29.88/0.8323	31.13/0.9361	31.77/0.8199	34.49/0.9588	31.02/0.9198	31.15/0.8808
	JPEG2K	22.10/0.6539	27.01/0.7674	26.27/0.7426	25.79/0.7323	24.86/0.8017	27.35/0.6482	25.52/0.7986	30.86/0.8397	25.97/0.6983	25.63/0.7649	28.27/0.7161	28.00/0.8183	26.33/0.8243	26.69/0.7543
	DIP	27.25/0.7498	30.92/0.8403	30.18/0.8692	30.79/0.9036	28.25/0.9091	29.62/0.7611	28.85/0.8817	31.81/0.8947	29.99/0.8333	31.18/0.9085	30.30/0.7782	32.22/0.8908	30.63/0.9203	30.31/0.8570
	DD	26.35/0.7029	24.27/0.7066	29.16/0.8508	28.80/0.8421	26.44/0.8932	25.99/0.7398	27.34/0.8634	34.87/0.9274	28.68/0.8073	30.02/0.9151	31.07/0.8003	33.09/0.9277	30.22/0.9052	29.22/0.8371
	ZS-N2N	28.63/0.7992	28.45/0.7803	32.08/0.9041	31.54/0.9141	28.70/0.9018	30.64/0.8048	29.67/0.8955	34.02/0.8817	31.63/0.8801	31.65/0.9120	31.28/0.8089	32.84/0.9009	31.27/0.9277	30.95/0.8701
	ZS-N2S	20.92/0.5844	21.40/0.5730	21.37/0.6468	20.78/0.4893	15.80/0.4987	22.03/0.5641	17.49/0.5154	8.43/0.3637	22.69/0.6725	12.21/0.6124	21.60/0.6438	11.58/0.4462	20.37/0.7871	18.18/0.5690
	S2S	22.36/0.5810	30.39/0.8769	22.74/0.7485	22.72/0.7108	17.44/0.7015	17.23/0.4383	17.41/0.7121	16.78/0.8102	15.65/0.4463	26.92/0.8838	24.20/0.7398	15.48/0.7528	14.84/0.5400	20.61/0.6879
	ZS-NCD	28.10/0.7831	33.85/0.9208	31.49/0.9051	32.65/0.9345	29.23/0.9355	30.51/0.7891	29.69/0.9077	35.85/0.9381	31.49/0.8867	33.21/0.9445	31.60/0.8281	35.07/0.9601	32.37/0.9448	31.93/0.8983
	Lottery Prior	29.36/0.8022	34.57/0.9234	32.91/0.9175	32.81/0.9253	31.21/0.9526	30.44/0.7456	31.66/0.9288	36.39/0.9479	32.26/0.8649	34.53/0.9589	32.07/0.8054	35.42/0.9707	32.02/0.9087	32.74/0.8963
	BM3D	26.56/0.6892	30.70/0.8824	28.06/0.7955	27.59/0.7664	25.90/0.8698	28.87/0.6978	26.93/0.8396	33.51/0.9087	27.35/0.7428	28.72/0.9056	30.11/0.7823	31.66/0.9324	28.61/0.8643	28.81/0.8213
25	JPEG2K	23.98/0.5711	24.47/0.6801	24.14/0.6400	23.53/0.5801	21.36/0.7014	21.70/0.5692	22.77/0.6809	28.30/0.7960	23.51/0.5667	23.03/0.7101	26.47/0.6729	25.23/0.7077	23.58/0.7389	24.32/0.6676
	DIP	25.70/0.6734	27.72/0.7021	27.79/0.7999	27.86/0.8126	25.11/0.8382	28.16/0.6596	26.03/0.8164	31.17/0.8384	27.27/0.7446	29.02/0.8785	28.82/0.7319	29.20/0.8106	28.65/0.8820	27.85/0.7837
	DD	25.56/0.6589	23.56/0.6676	26.94/0.7758	26.77/0.7542	24.76/0.8397	28.37/0.6819	25.72/0.8058	32.22/0.8849	27.09/0.7366	27.68/0.8566	29.42/0.7615	29.94/0.8937	28.11/0.8641	27.40/0.7832
	ZS-N2N	26.85/0.7287	26.73/0.7136	29.00/0.8282	28.30/0.8312	25.95/0.8462	28.71/0.7250	26.76/0.8248	31.31/0.8156	28.88/0.7940	29.03/0.8683	28.95/0.7282	29.78/0.8205	28.54/0.8768	28.36/0.8001
	ZS-N2S	19.22/0.4963	15.44/0.3973	22.05/0.5859	21.55/0.5581	16.64/0.5734	25.76/0.6423	17.98/0.6324	20.90/0.6792	22.30/0.6170	21.29/0.7778	22.45/0.6535	17.04/0.6502	22.40/0.7964	20.39/0.6200
	S2S	18.66/0.4947	24.97/0.7969	19.43/0.6054	20.69/0.5890	16.30/0.6168	15.08/0.3846	17.57/0.5267	15.53/0.7605	14.11/0.3892	21.34/0.7849	22.40/0.6795	13.60/0.6085	13.69/0.4890	17.95/0.5998
	ZS-NCD	26.54/0.7128	30.65/0.8388	28.78/0.8332	29.46/0.8718	26.83/0.8847	29.14/0.7367	27.05/0.8393	32.99/0.8870	28.83/0.8027	30.18/0.8917	29.62/0.7656	31.56/0.8898	29.63/0.9023	29.33/0.8351
	Lottery Prior	27.00/0.6891	32.60/0.9057	29.77/0.8381	29.66/0.8461	28.00/0.9080	28.82/0.6605	28.77/0.8765	33.04/0.9184	28.86/0.7658	31.54/0.9399	30.52/0.7776	32.08/0.9579	29.21/0.8542	30.00/0.8414

Table 18. Set13 denoising performance comparison under Poisson noise (PSNR and SSIM).

α	Method	Baboon	Barbara	Bridge	Coastguard	Comic	Face	Flowers	Foreman	Man	Monarch	Peppers	PPT3	Zebra	Average
15	BM3D	24.46/0.5651	26.63/0.7712	25.18/0.6462	24.45/0.5186	22.07/0.7101	27.46/0.6342	23.51/0.7017	29.05/0.7747	24.98/0.6405	25.60/0.8101	27.53/0.7041	26.16/0.7288	26.19/0.7802	25.64/0.6912
	JPEG2K	22.01/0.4873	21.82/0.5515	21.77/0.5163	21.69/0.3727	18.47/0.5237	24.97/0.5289	20.42/0.5841	23.83/0.6996	21.69/0.4723	20.21/0.5809	24.17/0.6012	23.70/0.5858	21.04/0.6381	21.76/0.5494
	DIP	24.30/0.5907	25.05/0.6010	25.46/0.7069	24.21/0.5996	21.79/0.7282	26.99/0.6150	23.41/0.7456	26.03/0.7686	25.10/0.6444	26.34/0.8136	26.26/0.6510	25.44/0.7088	26.27/0.8180	25.14/0.6916
	DD	23.89/0.5750	23.16/0.6479	24.83/0.6873	23.80/0.5544	21.52/0.7107	27.22/0.6387	23.47/0.7339	29.25/0.8522	25.03/0.6514	24.67/0.7745	27.09/0.7156	25.12/0.7646	25.43/0.8020	24.96/0.7006
	ZS-N2N	24.82/0.6330	25.17/0.5768	25.82/0.7281	25.35/0.6946	21.99/0.7055	27.11/0.6779	23.67/0.7490	27.20/0.6491	26.60/0.7290	25.99/0.7740	26.01/0.6747	25.02/0.6004	26.13/0.8190	25.37/0.6878
	ZS-N2S	21.39/0.5390	21.46/0.4084	22.20/0.6428	21.83/0.5714	15.53/0.5771	25.14/0.6103	17.97/0.5082	23.30/0.7854	22.94/0.5959	21.15/0.7157	21.53/0.5467	19.43/0.6122	22.08/0.7725	21.23/0.6066
	S2S	16.58/0.5492	17.66/0.6523	18.07/0.6269	17.50/0.5319	14.18/0.6548	18.87/0.6736	14.80/0.7562	14.80/0.7562	14.78/0.7188	17.73/0.7807	21.51/0.6626	12.61/0.5630	22.09/0.7728	19.23/0.6553
	ZS-NCD	24.66/0.5807	27.63/0.7844	26.43/0.7443	26.40/0.7436	23.17/0.7781	27.53/0.6150	24.30/0.7680	29.58/0.8302	26.15/0.6997	26.46/0.7991	27.34/0.7196	26.68/0.7217	27.45/0.8440	26.44/0.7434
	Lottery Prior	24.49/0.6127	27.43/0.8339	26.30/0.7431	26.08/0.6449	23.09/0.7903	27.99/0.6595	24.93/0.7745	26.66/0.8711	26.81/0.7064	27.14/0.8942	28.57/0.7543	24.71/0.9084	27.45/0.8240	26.35/0.7702
	BM3D	20.61/0.4799	21.75/0.5241	22.93/0.6748	22.04/0.5620	19.94/0.6637	24.88/0.6358	22.35/0.7387	22.34/0.5167	23.11/0.5790	21.99/0.6799	22.75/0.5533	20.01/0.4239	23.48/0.7582	22.17/0.5992
25	JPEG2K	22.59/0.5060	22.38/0.5732	22.10/0.5904	22.12/0.4455	19.59/0.6087	25.30/0.5533	21.51/0.6444	25.19/0.7343	22.10/0.4841	21.52/0.6226	24.78/0.6181	22.07/0.6515	22.44/0.7057	22.65/0.5952
	DIP	24.85/0.6243	24.02/0.5860	26.63/0.7646	25.41/0.6907	22.75/0.7640	27.42/0.6386	24.61/0.7935	29.42/0.8138	26.07/0.7119	27.02/0.8225	27.36/0.6839	26.86/0.7348	27.25/0.8670	26.13/0.7289
	DD	25.51/0.6017	23.30/0.6625	25.78/0.7389	24.96/0.6226	22.80/0.7680	28.18/0.6808	24.65/0.7872	30.34/0.8723	25.91/0.6824	26.31/0.8306	28.07/0.7282	26.92/0.7823	26.80/0.8309	26.04/0.7373
	ZS-N2N	24.81/0.6876	25.11/0.6387	27.17/0.7874	26.65/0.7522	23.45/0.7635	28.28/0.7277	24.59/0.8033	32.68/0.7031	28.10/0.7878	27.67/0.8276	27.47/0.6713	26.90/0.6864	27.43/0.8511	26.75/0.7555
	ZS-N2S	19.29/0.4409	21.36/0.5393	21.65/0.6879	20.44/0.4223	16.56/0.4988	25.41/0.6147	15.88/0.5870	22.50/0.8037	19.35/0.6591	21.09/0.6962	24.71/0.6801	22.47/0.7677	22.00/0.7806	21.19/0.6312
	S2S	17.40/0.5115	24.90/0.7441	18.60/0.6458	22.99/0.6967	15.48/0.5679	25.07/0.6826	19.74/0.7193	15.06/0.7385	26.12/0.7550	18.86/0.8094	23.26/0.6999	12.85/0.6302	22.89/0.8084	20.18/0.6922
	ZS-NCD	25.23/0.6145	29.09/0.8189	27.60/0.7969	27.66/0.8053	24.50/0.8223	28.31/0.6908	25.65/0.8231	30.99/0.8571	27.21/0.7491	27.82/0.8309	28.40/0.7464	27.98/0.7490	28.42/0.8712	27.60/0.7827
	Lottery Prior	25.50/0.6398	30.12/0.8641	27.68/0.7852	27.78/0.7689	24.90/0.8432	28.82/0.6810	26.51/0.8242	28.04/0.8894	28.04/0.7571	29.03/0.9118	29.61/0.7697	26.90/0.9220	28.83/0.8586	27.80/0.8088

Table 19. Kodak24 Denoising performance comparison under AWGN denoising (PSNR and SSIM).

	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Average
JPEG2K (15)	25.20/0.7129	25.23/0.6996	25.81/0.7678	25.00/0.7299	25.55/0.7372	26.71/0.7543	28.49/0.7880	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776	29.23/0.7776
BM3D (15)	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724	29.91/0.8724
DIP (15)	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886	26.10/0.7886
DD (15)	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750	23.89/0.5750
ZS-N2N (15)	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390	21.39/0.5390
ZS-N2S (15)	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492	16.58/0.5492
YSPNet (15)	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756
BMD15 (25)	26.36/0.7554	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112	27.54/0.8112
ZS-N2N (25)	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724	29.79/0.8724
ZS-N2S (25)	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429	26.30/0.6429
S2S (25)	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118	23.60/0.7118
YSPNet (25)	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756	23.30/0.6756